Electoral Effects of U.S. Soft Power: Evidence from Hollywood Movies in Cold War Italy *

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July 2026

Abstract

This paper examines whether U.S. movies, a form of soft power, shaped Italian elections during the Cold War. Using fixed effects and instrumental variables on a new panel dataset, we find that greater exposure to American films shifted votes from the Communist Party to the U.S.-backed Christian Democrats, with small effects on other parties and turnout. Analysis of film plots, dialogues, and contemporary surveys provides evidence consistent with a cultural transmission mechanism: whereas domestic films stressed collective themes and redistribution, Hollywood affected electoral choices by embedding values of individualism, agency, meritocracy, and markets into Italian popular culture.

Keywords: soft power, cinema, Cold War, voting, capitalism, communism

JEL Codes: D72, D74, F50, L82, N44

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1 Introduction

Soft power, as conceptualized by Nye (1990), refers to a nation's ability to influence others without force or financial incentives, typically through cultural exports such as music, literature, and cinema. The U.S. wielded soft power extensively throughout the 20th century, helping consolidate its leadership of the Western world (Nye, 2004). However, the effectiveness and returns of these investments are increasingly debated: the American government has recently moved to dismantle key instruments of its soft power apparatus, including the Agency for Global Media and the Agency for International Development (Exec. Order No. 14238, 2025; Exec. Order No. 14169, 2025). Whether such policies are well-founded depends on credible evidence regarding the returns to soft power, which existing empirical work has struggled to provide due to measurement and identification challenges.

In this paper, we study the relationship between American movies and electoral outcomes of foreign countries by examining the impact of exposure to Hollywood films on Italian elections from 1953 to 1972, a period marked by the Cold War. American movies represent a major export industry and accounted for a significant share of global movie revenue in that period (Fan et al., 2024), and scholars in film history have extensively documented their cultural influence on foreign societies (Wanger, 1950; Gundle and Guani, 1986; Brunetta, 1993; Treveri Gennari, 2011; Swann, 1991). Notably, Hollywood films often promoted Western capitalist values and culture, which can potentially shape foreign citizens' perspectives and political preferences.[1] The highly polarized political environment of Cold War Italy provides an ideal setting to study this question: the country featured the largest communist party in Western Europe, the Partito Comunista Italiano (PCI), on one side, and a conservative, U.S.-backed party, Democrazia Cristiana (DC), on the other.

We construct a novel panel dataset combining yearly information on cinema availability, U.S.

[1]Historically, totalitarian regimes recognized the persuasive powers of cinema early on and used it extensively as an instrument of internal propaganda. For instance, Hitler's National Socialist German Workers Party established a Film Department as early as 1930, and both Mussolini and Stalin had Film Departments in their cabinets. The ongoing debate on the ban of TikTok in the U.S. (Maheshwari and Holpuch, 2025) and the heavy regulation of foreign movie imports in China (Cieply and Barnes, 2013) illustrate how media continue to be viewed as powerful tools of influence by both democratic and non-democratic governments.

movie revenue shares, and electoral returns across Italian provinces over the five national elections between 1953 and 1972, drawing on multiple primary sources. Our main explanatory variable is constructed by interacting the local availability of movie theaters with the national share of revenues from U.S. movies (i.e., the ratio of total box office revenues from U.S. films to total movie revenues in Italy in that year) in the year preceding each election.

Identification raises two concerns. Both voters' choices and exposure to U.S. movies may be correlated with local characteristics, and both may trend over time, generating spurious correlations. We exploit the panel structure of our data through location and election fixed effects, which absorb time-invariant characteristics influencing both electoral and movie-going decisions (e.g., cultural traits), nationwide secular trends such as changing attitudes toward communism,[2] and election-specific shocks common across locations. Identification then comes from the remaining variation: exposure varies over time within each location and across locations within each election year. The historical context further allows us to rule out confounding from Soviet movies, whose penetration of the Italian market was negligible.

Our analysis shows that higher exposure to Hollywood movies increased the vote share of the DC, and decreased that of the PCI. According to our preferred specification, a one standard deviation increase in exposure in the year before the elections leads to an increase in vote share for the DC of 0.7 percentage points, and a decrease for the PCI of 0.4 percentage points. These effects correspond to changes of approximately 1.9% and 1.6%, or one-tenth and one-twentieth of a standard deviation, respectively. The effects are of the expected sign and statistically significant, but small in magnitude considering the high overall exposure to movies of the Italian public.[3] To benchmark this effect, we compute an annualized persuasion rate (DellaVigna and Gentzkow, 2010) of approximately 1.2% in our baseline specification (1.1% to 2.0% across IV estimates), an order of magnitude smaller than direct political propaganda but highly consequential given the massive

[2]These trends can be set off by historical events. For instance, 1956 was marked by Khrushchev's denunciation of Stalin's atrocities and cult of personality, and by the 1956 Hungarian and Polish revolutions, which were highly debated in Italy. This impacted the perception of communism in the Soviet Union's satellite states in Central and Eastern Europe, as well as in Western Europe.

[3]The average Italian went to the movies more than 1.25 times a month.

audience reach.[4] This finding is consistent with the fact that most U.S. movies did not carry any explicit political message and were tailored to a domestic audience. We do not find any meaningful effects on the vote share of other Italian parties.

We address two major identification concerns through an instrumental variable analysis that corroborates our baseline results. To account for the possibility that movies could be released in Italy strategically to influence electoral outcomes, we digitize information on the success of U.S. movies in France and use it to construct an instrumental variable. While the success of U.S. movies is strongly correlated across the two countries, France and Italy have major institutional differences, and their electoral calendars are not aligned.[5] A second concern is that access to movies could be manipulated by either domestic or foreign governments. Our instrument replaces contemporary local exposure to cinema with historical exposure measured before the implementation of the Marshall Plan and the establishment of the first Italian conservative government could meaningfully affect local cinema access. The IV estimates are consistent with those obtained from our fixed effects model and maintain strong statistical significance across all specifications. Further robustness checks show that our main results are robust to different econometric specifications, alternative measures of the explanatory variable, constructing the instrument using the Netherlands instead of France, standard errors accounting for spatial correlation, and a series of falsification tests. Finally, exposure to U.S. movies more than one year prior to an election has no independent effect on our main outcomes and does not alter our baseline estimates when used as a control.

To investigate the mechanisms underlying these effects, Section 6 examines whether Hollywood movies affected political preferences by shaping the values and attitudes of Italian viewers. We propose a cultural transmission mechanism: cultural products embed social and political values, and repeated exposure to these products can influence individuals' beliefs and preferences, which, in turn, affect electoral choices. We provide evidence for each link in this mechanism. First,

[4] We provide a detailed description of the methodology and the corresponding calculations underlying these persuasion rates in Appendix C.

[5] The historical context and cultural proximity of Italy and France explain the strength of the correlation we observe: Michalopoulos and Rauh (2024) discusses how a movie's success is related to its alignment with the core values and beliefs of its audience.

we document that American popular culture achieved substantial penetration in Italy during the postwar period, while Soviet cultural influence remained limited. Second, using large language models to analyze film narratives, we show that American movies systematically emphasized values associated with individualism, agency, meritocracy, and market-oriented views relative to Italian productions, which placed greater emphasis on collective themes and redistribution. We further show that the language used in American movies was closer to that of conservative parliamentary discourse than that of Italian movies, suggesting that Hollywood content was ideologically more aligned with the political platform of the DC. Finally, using survey evidence on Italians' political attitudes, we show that individuals with more favorable views toward the U.S. and greater alignment with its foreign policy preferences were also more likely to support the DC over the PCI. Taken together, these results suggest that Hollywood movies influenced electoral outcomes not through direct political persuasion, but by gradually shaping cultural values and geopolitical preferences in a direction consistent with the DC's political agenda.

Finally, we examine the persuasive power of movies in different areas and identify suggestive patterns of heterogeneity according to locations' historical political orientation. Specifically, we find that the increase in support for the DC is driven by areas where the DC was historically weaker, indicating the persuasive effect of U.S. movies. Additionally, we observe the PCI experiencing losses primarily outside of the most urban areas, represented by the Province Capitals. However, we do not find significant differences in effects across other geographical dimensions.

Related Literature. We contribute to the existing discussion on soft power, providing the first causal estimates of its effects on electoral results, to the best of our knowledge. Previous quantitative work has linked trade tariffs to measures of soft power (Fan et al., 2024) and international trade to political alignment across countries (Kleinman et al., 2024). We address the relation between a suggested factor in the production function of soft power, cinema, and one of the main goals of American foreign policy in the period, installing friendly governments in foreign countries.

Our analysis of the impact of exposure to American movies in Italy links our study to others that also empirically examine the effects of the influence of the U.S. on foreign countries during the Cold War. This literature has focused on interventions with a clear and direct involvement by the U.S. government: Dube et al. (2011) examine the impact of U.S.-backed coups on stock prices of U.S. companies in Iran, Guatemala, Cuba, and Chile; Berger et al. (2013) study the impact of CIA interventions during the Cold War on bilateral trade relations between the U.S. and foreign countries; Nunn and Qian (2014a) analyzes the determinants of food aid and its impact on conflict in recipient countries. In contrast, our analysis focuses on a subtler and undirected form of influence.

Our paper also contributes to the literature studying the causal effects of media on voting choices (Gentzkow, 2006; Enikolopov et al., 2011; Adena et al., 2015; Campante et al., 2018; Durante et al., 2019; Enikolopov et al., 2020; Xiong, 2021; Manacorda et al., 2025). To the best of our knowledge, ours is the first study to document the causal impact of cinema in determining foreign electoral outcomes.[6] Key differences with previous work on the effects of foreign media are our focus on electoral outcomes and the intentionality of exporting movies as opposed to unintentional exports such as the case studied in DellaVigna et al. (2014) where Serbian radio signal bled through borders increasing anti-Serbian sentiments among Croats.[7]

The paper proceeds as follows. In the next section, we provide the relevant historical background. In Section 3 we describe the data. In Section 4 we explain our empirical strategy. Section 5 presents the results of our analysis. Section 6 provides evidence for the cultural transmission mechanism. Section 7 concludes the paper.

[6] Dahl and DellaVigna (2009) study the short-run effects of movie violence on crime. Jacobsen (2011) shows that exposure to Al Gore's documentary, raising awareness on climate change, caused an increase in the purchase of voluntary carbon offsets. Ang (2023) examines the impact of exposure to the movie "The Birth of a Nation" on racial hate, while Esposito et al. (2023) explores its effects on patriotism and public discourse. Within the U.S. context, Ren Tan and Wang (2024) provides suggestive evidence that the shift in film content caused by the Hollywood Red Scare increased Republican support.

[7] Kern and Hainmueller (2009) and Bursztyn and Cantoni (2016) study the foreign influence on former East Germany through differential access to Western German television during the communist regime. Our analysis differs from theirs in at least two respects: the focus on a democratic country as opposed to a dictatorship and the use of electoral outcomes instead of survey measures of regime support or consumption patterns.

2 Historical Background

In this section, we discuss the Cold War historical context, the political scenario in Italy in the period 1945-1973, and the state of the movie industry in Italy. The historical accounts lend credibility to the idea that, during this time, cinema was a salient way to learn about the two main ideologies competing on the world stage: capitalism and communism.

Context During the Cold War, the hostility between the U.S. and the Soviet Union was not only a military and political rivalry, but also a contest between two superpowers vying to promote their respective ideologies on a global scale. The Soviet Union aimed to spread communism worldwide by fostering relationships and supporting communist parties in foreign countries. Similarly, the U.S. government sought to promote capitalist ideals and democratic values, while deeply concerned by the growth of communist parties in Western European democracies. To counter this threat, the U.S. government employed both overt and covert operations to attract countries within its sphere of influence.[8] In March 1948, the National Security Council report 1/3 directed the U.S. to *"immediately undertake further measures designed to prevent the Communists from winning participation in the government"* and to *"continue to assist the Christian Democrats and other selected anti-Communist parties"* (NSC 1/3, 1948). In 1959, "Operations Plan for Italy" prescribed that *"we should not support a government which depends for its majority on either the Communist Party, or Socialist Party as presently oriented"* (Operations Coordinating Board, 1959). The response to National Security Study Memorandum 88 in 1970 defined U.S. goals as *"a democratic Italy without either Communist or neo-Fascist participation in the government"* (NSSM 88, 1970).

The U.S. government's interventions were not the only channel through which foreign countries were influenced. Hollywood, through its movies, was a clear testimony of the cultural and economic expansion of the U.S., exercising a significant influence worldwide. American movies

[8]The aid provided under the Marshall Plan and the CIA interventions are perhaps the most striking examples of this strategy (Berger et al., 2013). A declassified Office of the Secretary of Defense study notes that Ambassador Clare Boothe Luce's "biggest tool was the CIA's political action program," and that a 1956 agency review found "at least 90% of its expenditures in Italy went for covert political, psychological, and propaganda operations" (Office of the Secretary of Defense, Historical Office, n.d.).

generated sentiments of admiration towards the U.S., the American way of life, and the American model of democracy. Hollywood depicted a wealthy society where opportunities to climb the social ladder were real and where everyone could enjoy a high standard of living, especially in comparison to contemporary European standards.[9] A 1951 State Department information-program annex (NSC 114/1, Annex 5) stated that *"the Department of State was to underwrite losses incurred in the production of desirable films by Hollywood companies for commercial use abroad"* (NSC 114/1, 1951).

The conjecture of positive effects of U.S. movies on perceptions of America is reinforced by the fact that communist politicians were displeased by the American cultural influence, acknowledging the potentially harmful effects of U.S. movies.[10] At the same time, according to other contemporary commentators, not all movies depicted the U.S. in positive terms. These observers focused on other parts of the message that U.S. movies could transmit: excessive materialism, inequality, and violence.[11]

Italian Politics After Italian Fascism and WWII ended, a provisional government composed of all major political parties was installed, and the monarchy was abolished by popular referendum in 1946. The new republican constitution went into effect on January 1, 1948, and the first free nationwide democratic election was held on April 18, 1948. Until 1994, the electoral law was based on proportional representation, entailing that each political party received a number of seats in proportion to the number of votes it received. The electoral law prescribed elections to be held

[9] As the New York Times wrote in 1955: *"Millions of people in this world know nothing of the United States except what they see in movies. Hollywood is the looking-glass in which is reflected the American way of life, its philosophy and ideology. [...] The incredibly high standard of living mirrored by our movies is a source of envy and incredulity in many lands. [People think that] in America even the tramps have cars."*

[10] *"Americans not only send their spies, soldiers and saboteurs, but also flood our country with books and movies which confuse and lure our people"*, Pietro Secchia, prominent member of the Italian Communist Party (PCI), on February 5, 1948 in *L'Unitá*, the PCI newspaper.

[11] The National Archives in Washington DC contain various letters of U.S. embassy staff trying to obtain the withdrawal of specific films from their country of assignment, due to their potential harm to perceptions of the U.S. For instance, in 1953, William R. Auman, Public Affairs Officer in Oslo, wrote that he was pleased that the Norwegian film censor had banned the Marlon Brando film, *"The Wild One"* as it *"presented practically all the standard misconceptions about America that our enemies stress such as claims that we are uncultured, rude, bombastic, impatient, lawless and addicted to mob violence. Films of this type when exported are harmful by presenting false or distorted impressions of American life as a whole."*

every five years, unless the Parliament chambers were dismissed earlier.[12]

The period we analyze is characterized by intense competition between Christian Democracy (DC) and the Italian Communist Party (PCI). From 1946 to 1992, DC was always the leader of a ruling coalition with other small centrist parties and, starting in 1963, also with the Socialist Party.[13] Since its creation, the DC emphasized family values and endorsed capitalistic ideals such as defense of free enterprise and private property (Ginsborg, 2003). PCI was the largest and best-organized communist party in Western Europe with close ties to Moscow, and its political and social influence continued to grow during the post-war decades. However, the PCI was de facto excluded from participating in the government through the political tactic known as the *"conventio ad excludendum"* (Colarizi, 2016). This was a U.S.-supported agreement among the major Italian political parties to exclude the PCI from the government, which was considered a threat to the established political order in Italy.

In 1973, after the *"Historic Compromise"*, the ideological divide between the two parties decreased. Enrico Berlinguer, General Secretary of the PCI, proposed to set aside the hostility with the DC and started to distance the party from the Soviet Union both in terms of relationships and of ideology. As part of this strategy, the *"conventio ad excludendum"* was dissolved and the DC came to govern with the tacit support of the PCI from 1976 to 1979 (Colarizi, 2016).

Movie Industry During the 1930s, the Italian movie industry had been heavily protected and financed by the fascist state. In an effort to reduce the current account deficit and to support national production, the Fascist government enacted the *"Alfieri Law"* in 1938 (regio decreto-legge del 4 settembre 1938, n. 1389), imposing an embargo on foreign movies (Venturini, 2012, p.142-147). The decree was strictly enforced: the number of American movies imported went from 162 in 1938 to 0 in 1940 (Brunetta, 1993). After WWII ended, the ban was lifted and the distribution of foreign movies was allowed again. At that time, the Italian film industry was in a state of

[12] Although Italy was characterized by high political instability of government, with 22 different government coalitions between 1953 and 1972, early parliamentary elections did not occur until 1972.

[13] Appendix Table A5 reports results from all elections in our sample. DC and PCI were the dominant political actors, with a combined share of total votes above 60% in each election, with considerable distance from the third party.

considerable devastation and Italian studios were unable to compete with Hollywood.[14] Given the difficulties and the unsatisfied demand for American movies, Hollywood came to dominate the Italian movie industry, unloading a six-year reserve of unseen American movies on an eager Italian public.[15] During the 1950s, Italy had the highest number of cinemas (11,641 compared to 5,806 of France and 6,885 of Germany) in Western Europe, making the medium the most popular form of entertainment for Italians (Treveri Gennari, 2011). Italy has a strong tradition in dubbing, so that American movies were shown in Italian without subtitles, allowing even illiterate people to watch Hollywood productions.

Soft Power Soft power refers to a country or organization's ability to influence others through non-coercive means such as culture, diplomacy, and ideology. Soft power manifests in various forms, ranging from the influential power of books to the role of sports in promoting cross-cultural understanding and cooperation (Nye, 1990).

The publication of books critical of the Soviet Union such as *"Darkness at Noon"* and *"Animal Farm"* during the Cold War played a key role in shaping public opinion about the nature of the Soviet regime. In music, the U.S. promoted rock and roll as a symbol of freedom and individualism, while the Soviet Union used classical music to showcase its cultural superiority. Soft power has also played a significant role in the realm of sports (Watanabe and McConnell, 2008). For example, the "ping-pong diplomacy" of the 1970s brought American table tennis players to China to build bridges between the two nations. The U.S. higher education system has also been described as an instrument of soft power, drawing in scholars and students from across the globe (Nye, 2005).

More recently, China's rise as a global power has been accompanied by a concerted effort to expand its influence through the use of soft power (Zheng and Zhang, 2012). Through cultural outreach, economic investment, and diplomatic engagement, China has sought to promote its values

[14] Many large film studios had been plundered by the fascists and the Germans; Cinecittá, the largest center for movie production in Europe before WWII, was used as a refugee camp until 1950 (Steimatsky, 2009). The Italian film industry was fragmented and unable to consolidate: in 1948, 87 percent of the Italian production was carried out by studios that only produced one movie before disappearing from the market; in 1950, only three studios produced more than one movie (Brunetta, 1993).

[15] Examples of classics not imported due to the blockade and released after the war include *"The Wizard of Oz"* and *"Gone with the Wind"*.

and interests on the world stage.

3 Data

This section describes the data sources used in the analysis and presents descriptive statistics and patterns in the data. First, we present the measures of exposure to cinema, which form the basis of the main explanatory variable. Second, we introduce the electoral data used to construct political outcomes, followed by census data providing demographic and socioeconomic controls. Finally, we describe the data employed to examine underlying mechanisms, including contemporary survey data and newly constructed textual datasets on movie content.

3.1 Data Sources

Access to cinema and movie revenues in Italy. We obtain information on cinema availability from previously unexplored records digitized by the *Italian Society of Authors and Publishers* (SIAE), Italy's state-mandated monopolist for copyright intermediation and royalty collection.[16] SIAE publishes yearly reports called "Annuari dello Spettacolo", the most detailed quantitative representation of the conditions of the Italian entertainment industry.[17] From SIAE, we obtain yearly data from the earliest available date, 1948, to 1973 on the number of movie theaters and the number of tickets sold at the *location* level, reported separately for each provincial capital and for the remainder of its province.[18] Accordingly, the unit of observation in our analysis is a *location*,

[16]Law 633/1941 (Art. 180) reserves copyright intermediation (including cinematographic reproduction) exclusively to the statutory collecting society (E.I.D.A./SIAE). Cinema tickets were issued in a single format bearing the SIAE stamp (*contrassegno SIAE*), and exhibitors had to compile SIAE-marked box office forms (*borderò*) recording admissions and receipts (Law 958/1949, Art. 26). The Italian National Institute of Statistics, ISTAT, characterizes SIAE cinema statistics as a nationwide, total-coverage collection, through SIAE's local offices, based on these administrative tax records for paid-admission events (Istituto Nazionale di Statistica (ISTAT), 2016).

[17]SIAE data from this period have been used extensively in the literature on postwar Italian cinema (Treveri Gennari and Sedgwick, 2015; Miskell and Nicoli, 2016; Sedgwick et al., 2019). Miskell and Nicoli (2016) note that the implied scale of commercial cinema venues in Italy (around 11-12k in the late 1950s) is consistent with an independent industry market survey (Motion Picture industry association materials) reporting 11,148 "motion picture theatres" in Italy in 1957.

[18]Provinces are Italian administrative divisions comparable to U.S. counties corresponding to NUTS3 Eurostat classification of territorial units. Our dataset is composed of 178 panel observations. For each of the 89 provinces, we observe information for the main municipality (*capoluogo di provincia*) and, separately, for the rest of the province.

defined as either (i) the main city of a province or (ii) the rest of the province excluding the main city. This structure is dictated by data availability, as the SIAE reports provide this as the finest spatial disaggregation consistently available over our period, allowing us to maximize the number of observations while remaining aligned with the underlying data-generating process. From SIAE, we also obtain yearly box office revenues and the number of movies displayed in Italy at the national level by country of production. Figure 1 provides a graphical representation of the location definition and displays the density of movie theaters in each of them.

Success of U.S. movies abroad. We measure the success of U.S. movies at the box office in other nations for the period 1951-1973 using data from Gyory and Glas (1992). From this source, we obtain the yearly share of revenues of U.S.-produced movies in movie theaters in France, US_t^{FR}, and the Netherlands, US_t^{NL}. Our focus on France and the Netherlands is primarily driven by data availability, as these are the countries for which the information is most complete. By contrast, data for Germany begin only in 1955, whereas the first election included in our analysis takes place in 1953. While both France and the Netherlands are geographically close to Italy, are among the six founding member states of the European Union, and have similarly functioning democratic institutions during the same period, France shares important demographic and ethnolinguistic similarities with Italy that make us prefer it for the instrumental variable analysis. We consider the Netherlands as an additional robustness check to our main analysis.

National Elections. Electoral data on national elections are obtained from the Italian Ministry of the Interior. For each election, we observe the number of citizens eligible to vote, of citizens who cast a ballot, and of votes for each party in each municipality. Our main dependent variables are the share of votes for DC and PCI, computed as the number of votes gained in each location, divided by the total number of valid votes in that location. All electoral data are aggregated to the *location* level, as defined previously, so that each observation corresponds either to the main city

Data on cinema availability for the cities and related provinces of Gorizia and Trieste are missing until 1954, so we exclude them from our sample.

of a province or to the rest of the province excluding the main city. Since new provinces were instituted between 1945 and 1972, we construct our unit of observation using 1951 borders.[19] We focus on the five elections for the lower chamber from 1953 to 1972. The choice to stop at the 1972 election is driven by the historical considerations discussed in Section 2. The election of 1948 is excluded for several reasons: first, there is evidence that the American government attempted to influence the elections directly (Miller, 1983); second, the PCI did not run independently but as part of a larger coalition including the Socialists; last, not all the relevant variables are available for our access-to-movies measures.

Census Data. To account for observable characteristics that may be correlated with both electoral outcomes and exposure to Hollywood movies, we include time-varying demographic and economic variables aggregated at the *location* level, obtained from decennial Italian Censuses (1951, 1961, and 1971).

Film Data. To investigate the mechanism through which exposure to Hollywood movies may have influenced Italian voters, we construct a dataset of film summaries and dialogue from movies shown in Italian cinemas during our study period, which we detail below. This allows us to systematically compare the thematic content and linguistic patterns of American versus Italian films. We identify films using the HitParade Italia box office rankings, which record the top 100 films by ticket sales for each cinema season from 1954–55 through 1973–74.[20] We match these titles to the TMDB database to obtain standardized identifiers and country of origin,[21] classifying films as "American" if produced in the U.S. and "Italian" if produced in Italy (excluding U.S. co-productions from the sample).

[19]We use information from SISTAT (Historical System of Territorial Administration), providing information on the composition of Italian provinces starting in 1861.

[20]`https://www.hitparadeitalia.it/bof/boi/`

[21]TMDB (The Movie Database) is a community-curated online database of film and television metadata, including production-country information. See TMDB.

Plot Summaries. To analyze the value orientations embedded in film narratives, we obtain plot summaries from IMDb for films in our box office sample. We match films by IMDb identifier (derived from TMDB metadata) and retrieve all available plot summaries, keeping the longest summary per title.[22] Of the 2,000 films in our box office sample, 1,929 have an IMDb plot summary available, corresponding to a 96% coverage rate of the full sample. The annotated sample comprises 750 American, 897 Italian, and 276 films from other countries. Plot summaries are then annotated using a large language model as described in Section 6.2 and Appendix B.1.

Film Dialogues. As a proxy for film dialogue, we obtain English-language subtitles from the OpenSubtitles and Podnapisi databases.[23] We use English subtitles for *all* films – both American and Italian productions. This design choice ensures comparability: we measure differences in thematic content rather than detecting language-specific vocabulary.[24] Second, it avoids noise from language misclassification. Manual inspection revealed that non-English subtitles are sometimes mislabeled, so restricting to English minimizes this source of error. After matching and quality filtering, we obtain subtitle text for 1,109 films, with similar coverage rates for American (58%) and Italian (52%) productions. For analyses based on dialogue content, we split each film's subtitles into dialogue chunks.[25]

Parliamentary Speeches. To construct a measure of political orientation grounded in the Italian context, we use the ItaParlCorpus (Cova, 2025), a comprehensive annotated corpus of parliamen-

[22]IMDb plot summaries are crowdsourced. Registered users submit short summaries and longer synopses, which pass through IMDb's editorial review process before being published. TMDB (The Movie Database) provides standardized metadata including country of origin and cross-references to IMDb identifiers.

[23]We use OpenSubtitles because it is the largest publicly accessible subtitle repository, maximizing coverage across titles and years. We also draw on Podnapisi, which provides complementary subtitle coverage and improves retrieval when OpenSubtitles is missing or incomplete for particular titles. We thank the OpenSubtitles team for sending us the bulk data for the movies in sample.

[24]A potential concern is that translation may distort thematic content; however, we anticipate that any systematic translation bias would work against our findings as translators might try to smooth cultural differences.

[25]A dialogue chunk is a contiguous block of subtitle segments aggregated to roughly 800–1,600 characters (about 10–15 lines of dialogue). Aggregation respects the timing of the subtitle track: a new chunk starts whenever there is a silence of at least 30 seconds or the running block has lasted more than two minutes; otherwise, once the minimum length is reached, the chunker closes on the next sentence boundary. This unit is long enough to carry coherent thematic content while preserving local context, and gives us 56,505 chunks across our U.S. and Italian films.

tary debates from the Italian Camera dei Deputati covering 1948–2022. We extract speeches from 1948-1974, labeled by speaker party affiliation, which allows us to learn what linguistic patterns distinguish conservative parties (DC and other conservative/right-wing parties) from left-wing parties (PCI and far-left groups).[26] After splitting speeches into sentence-level segments and excluding centrist parties, we obtain approximately 293,000 speech segments.

Party Manifestos. We collect electoral manifestos from the Manifesto Project (Lehmann et al., 2024), specifically those published by the DC and PCI for elections between 1948 and 1979. These documents are used to validate the political alignment measure in Section 6.3.

ITANES survey data. We also obtain information on voters' attitudes towards the U.S. and political preferences using individual-level data from the 1968 and 1972 waves of the Italian National Election Study (ITANES), a series of electoral surveys conducted on a representative sample of the Italian population in every election year starting in 1968. The data complements electoral data at the province-level by eliciting feelings towards the U.S. government, Americans, Russians, the DC, and the PCI through direct questions. The survey also contains important individual-level characteristics on sex, age, education, and income of the respondents, which we use as controls. The data contain the location of the respondents so that we can match them with information on cinema availability. We provide details on the construction of the variables in Appendix D.

3.2 Descriptive statistics

In Table 1 we report summary statistics on the main variables in our sample.[27] For each unit of observation, the mean number of movie theaters in the period 1950-1972 was 56, corresponding to 2.04 movie theaters per 10,000 inhabitants on average, ranging from 1 in Nuoro to 466 in the

[26] See Section 6.3 for the full list of parties in each category.

[27] Table A6 displays the corresponding statistics by Province Capital and Rest of the Province. It shows that Province Capitals are on average about half as populous as the Rest of the Province, but have lower employment in agriculture, higher employment in the services sector, and a higher share of people with a college degree. They also tend to have higher DC vote shares and lower PCI vote shares; as displayed in Figure 1, Province Capitals have a lower number of movie theaters relative to their population, reflecting their higher population density.

province of Milan in 1966. The number of movie theaters in 1948 was lower on average than in subsequent years. The statistics on the yearly number of movie shows by country of production paint a clear picture of the importance of the Hollywood movie industry in the Italian market and the negligible penetration of Soviet movies.

The left panel of Figure 1 plots the distribution of movie theaters per 10,000 inhabitants in 1948 across Italy. The distribution presents considerable variation across the Italian peninsula, with the highest concentration of movie theaters found in the Northern Italian provinces and the lowest in the South. The right panel of Figure 1 provides information on the content of the movies shown in movie theaters, presenting the yearly distribution of U.S. movies by genre.[28] A visual inspection reveals that Drama and Romance was the most successful genre throughout the time period, Crime and Thriller movies grew in importance over time, while War movies decreased their share.[29] History and Cinema Studies scholars have produced lists of movies identified as anti-Communist,[30] such as *"Man on a Tightrope"* or *"The Commies Are Coming, the Commies Are Coming."* We analyze movie content and its possible effect channels on electoral choices in Section 6.

4 Empirical Strategy

We study the effects of exposure to U.S. movies on Italian national elections during the Cold War using two main approaches. First, we focus on variation within municipalities over time while controlling for common (national) trends in the variables of interest. This is obtained by taking advantage of the panel structure of the dataset and estimating the following equation with OLS:

[28] We do not possess information on which U.S. movies were shown in Italian movie theaters and where they were displayed. The data provides information on U.S. movies by year of production commercialized in the U.S.

[29] Examples of Drama and Romance movies are blockbusters such as *"The Graduate"* (1968) and *"Love Story"* (1970). It is worth noting that movies revolving around sports like *"Follow the Sun"* (1951) or *"The Square Jungle"* (1956) have been identified as emphasizing capitalistic and individualistic messages (Shaw and Youngblood, 2017).

[30] See Shaw and Youngblood (2014), for a discussion of how movies were used by both Soviets and Americans for propaganda, and the University of Wisconsin Library Guide at https://guides.lib.uw.edu/c.php?g=341346&p=2303736 for a list of Hollywood movies related to the U.S. government anti-Communist efforts.

$$PartyShare_{i,t} = \beta ExposureUS_{i,t-1} + c\mathbf{X_{i,t}} + \theta_i + \alpha_t + \nu_{i,t} \tag{1}$$

where $PartyShare_{i,t}$ denotes the vote share of either DC or PCI; i indexes the cross-sectional dimension of the panel, i.e., Province Capital and Rest of the Province; and t indexes time, i.e., election years.

The coefficient of interest β captures the effect of differential exposure to U.S. movies before elections on the vote share of the two major parties. $ExposureUS_{i,t-1}$ is a standardized variable measuring exposure to U.S. movies in location i in year *t-1*, the year before the elections. We use the year before elections since all elections in our sample are held between March and May, while box office data refer to calendar years. Focusing on the success of Hollywood movies in the year before the election allows us to capture the short-term effect of U.S. movies; in Section 5, we show that effects fade rapidly.[31]

Since local yearly box office data by country of production is not available, we construct our exposure measure as $ExposureUS_{i,t-1} = US_{t-1} \times Cinema_{i,t-1}$, where US_{t-1} denotes the share of total box office revenues in Italy in year *t-1* that is attributable to U.S.-produced movies (i.e., total revenues from U.S. films divided by total movie revenues in Italy in that year), and $Cinema_{i,t-1}$ captures local access to cinema in location i at *t-1*. In our main specifications, we measure access to cinema using the number of movie theaters per capita in each location. As a robustness check, we also use an alternative proxy for cinema access, $Tickets_{i,t-1}$, defined as the number of tickets sold per capita. The interaction of year-to-year fluctuations in the appeal of Hollywood's output with cross-sectional differences in cinema infrastructure generates differential exposure to American movies across locations and elections.[32]

[31] As a robustness check, we also report estimates controlling for same-year exposure in Appendix Table A9 and discuss them in Appendix A.

[32] Our setting does not map onto the canonical staggered adoption difference-in-differences framework (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfoeuille, 2020). Those estimators are designed for settings with binary, absorbing treatment at staggered dates, whereas in our setting all locations are exposed throughout the sample period and exposure intensity varies continuously, rising and falling across elections. Estimators allowing for continuous treatment variables and no never-treated units (see, e.g., de Chaisemartin et al., 2024) still require units whose treatment is infinitesimally small, which are unavailable in our setting.

The panel structure of our data allows us to further control for time-invariant characteristics, in addition to some important time-varying ones. Parameter θ_i is a location fixed effect, capturing time invariant characteristics pertaining to i, and α_t are election year fixed effects, absorbing variation in the outcome common to all locations over time.[33] Because our treatment is the interaction of a time-varying variable with a location-time one, confounders would have to be trend variables that interact differentially with localities along the contours of our variables. The time trend would have to co-move with the U.S. movie share and affect localities differentially along something that co-varies with cinema exposure.[34]

The second approach builds on the previous design but constructs an instrument for cinema exposure that replaces contemporary access to movies with historical access (measured in 1948), and the success of Hollywood movies in Italy with the success of U.S. movies in France. Appendix Figure A3 provides graphical evidence on the relevance of our instrument for both components of the instrument separately. With a continuous treatment and instrument, two-stage least squares recovers a weighted average of causal responses across exposed locations (Angrist and Imbens, 1995; Angrist et al., 2000), which in our setting places more weight on locations with denser 1948 cinema infrastructure.

This approach is robust to the potential concern of endogeneity bias caused by strategic behavior of a sophisticated government (Italian or American) that manipulates exposure to Hollywood movies. The instrument addresses the concern that the government might have promoted the construction of movie theaters in selected areas by fixing their number to 1948 levels, before a single-party government could influence it.[35] Additionally, replacing the Italian share of U.S. movies with the corresponding share in France in the instrument removes the possibility that American

[33] Year fixed effects absorb any component of the effect of U.S. movies that operates uniformly across localities. For example, a year of particularly persuasive Hollywood output may shift DC support across all of Italy, but this common shift is netted out by α_t. Our estimate therefore recovers only the differential response across locations with different levels of cinema access, which likely attenuates the estimated magnitude of the aggregate persuasive impact of U.S. movies.

[34] In Section 5, among other robustness checks, we show that the key findings are robust to controlling for location specific linear time trends and other covariates.

[35] Post-war governments prior to 1948 were supported by a large majority of parties and included ministries from both DC and PCI. Also, post-war reconstruction started after the enactment of the European Recovery Program (also known as the Marshall Plan) in 1948.

movie penetration in the Italian market was strategically increased to influence election outcomes. Instrumenting exposure can also reduce the attenuation bias coming from noisy measurement of local exposure to cinema or of movie success. However, the instrument may also discard some informative signal (e.g., the part of U.S. movie appeal that is not shared by the Italian and French market) and is expected to produce noisier estimates.

Our model for the second stage is almost unchanged compared to Equation 1, only substituting exposure with predicted exposure, while the first stage is:

$$ExposureUS_{i,t} = \gamma Instrument_{i,t} + c^{fs}\mathbf{X_{i,t}} + \theta_i^{fs} + \alpha_t^{fs} + \varepsilon_{i,t} \tag{2}$$

where $Instrument_{i,t} = US_t^{FR} \times Cinema_{i,1948}$.

For estimates of model 1 and of the instrumental variable approach we report standard errors clustered at the location level.

5 Empirical Results

Main Results. Table 2 shows the estimates of Equation 1 for the effects of exposure to U.S. movies on vote shares from the DC and the PCI. All specifications include election year and location fixed effects. In columns (1)-(4), we report results on the DC vote share. The coefficient of interest is positive and statistically significant at the 1% level in all specifications. It ranges from 1.2 percentage points (p.p.) in column (1) to 0.75 p.p. as we add more socio-economic controls. The smallest estimate, 0.75 p.p., represents a 2% increase relative to a baseline vote share for the DC of 37.7 p.p. In columns (5)-(8), we report results on the PCI vote share. The coefficient of interest is negative and statistically significant at the 1% level in column (5) with no controls, becoming smaller and maintaining 5% statistical significance as we add more controls. The magnitude of the negative coefficient, ranging from 0.9 p.p. to 0.4 p.p. from a baseline mean of the PCI vote share of 25.3 p.p., is statistically indistinguishable from the positive one on DC's shares. Overall, the OLS results show that areas with a greater concentration of movie theaters in years when U.S. movies were particularly successful before elections tend to vote more for the U.S.-endorsed party and less for the Soviet-supported party.

Since the electoral law in Italy was proportional, the effects that we find translate, according to the most demanding specification, into a gain of 4.72 elected members for the Chamber of Deputies for the DC and a 2.77 loss for the PCI out of a total of 630 parliamentarians. Hence, although these are sizeable, they do not have a groundbreaking impact. These effects could have made a more substantial difference had Italy had a majoritarian law, an electoral system where small differences in electoral outcomes can cause large differences in terms of real political power.[36]

IV and Robustness. Table 3 reports estimates obtained using the instrumental variable approach. The first column shows the relevance of our instrument by reporting the first-stage coefficient, which is positive, large in magnitude, and precisely estimated. The F-stat for the model without controls is 53 and remains above 28 as we add more controls. In columns (2)-(9), we report the second-stage coefficients. The IV estimates are of the same sign and of larger magnitude than the OLS estimates, maintaining strong statistical significance in all specifications.[37] These estimates confirm the finding that exposure to U.S. movies increased the share of votes for the DC and produced a similar decrease for the PCI. Since this approach is robust to potential strategic manipulation by the government, which is likely to bias upwards the ATE for the DC share,[38] these estimates suggest that such manipulation does not explain the main results.

In Table 4, we report results showing that our findings are robust to a variety of different specifications, measurements of the explanatory variable, and instrument choice. In Panel A, the depen-

[36]The effects that we find could have also caused a more meaningful difference had the coalition of parties led by the DC marginally reached 50% in 1953. In fact, only for the 1953 elections, the electoral law introduced a bonus system, the so-called *Legge Truffa*, which awarded extra seats to the coalition of parties that received 50% or more of the vote. However, the coalition of parties led by the DC obtained 49.8% of the vote and therefore the law was not applied.

[37]The two approaches have different estimands: with a continuous treatment and instrument, 2SLS recovers a weighted average of causal responses that places more weight on locations with denser 1948 cinema infrastructure (Angrist and Imbens, 1995; Angrist et al., 2000). The IV approach answers a narrower question than the preferred OLS, identifying the effect of exposure in locations where cinema attendance was historically greatest. For our central object of interest, how exposure to Hollywood reshaped aggregate electoral outcomes across Italy, the population-wide average captured by OLS is the more relevant quantity.

[38]In a proportional electoral system like the Italian system over the studied period, a strategic government would want to target exposure in highly responsive areas, thereby generating upward bias.

dent variable is the DC vote share, while in Panel B, the dependent variable is the PCI vote share. For both panels, columns (1)-(6) use the number of movie theaters per capita to measure access for each location, while columns (7)-(12) use the number of tickets sold per capita as an alternative measure. As an additional robustness check for the IV specifications, we also report results replacing the success of U.S. movies in France in the instrument with the success in the Netherlands in columns (5), (6), (11), and (12).[39] In even columns, we alter the specification of interest by adding the uninteracted local exposure to cinema (movie theaters or tickets per capita), which we label as contemporaneous exposure.[40] All reported specifications include the full set of controls, i.e., sectoral employment, labor participation, population, and education controls. The pattern of the estimates obtained is clear and shows that our results are robust to different specifications, measurements of the explanatory variable, and instruments. Comparing the coefficient in column (1) from our preferred specification, we observe that estimated coefficients are of the same sign in all cases, of similar magnitude in most cases, and largely maintain statistical significance.

Appendix A.1 collects additional robustness checks. First, we test including the treatment variable measured in different years. The effect only appears for exposure measured in the year before the election, fades two years out, and exposure measured in the election year itself – which largely reflects post-election months – is roughly half the size and statistically insignificant. Second, our conclusions are unchanged when standard errors allow for spatial correlation (Conley, 1999). Additionally, a permutation-based falsification test that randomly reassigns both components of the exposure measure places our estimates in the extreme tails of the resulting placebo distributions.

Finally, we investigate whether exposure to U.S. movies before elections had effects on the vote shares of other parties and on turnout.[41] Table 5 presents the results. We observe that turnout

[39]The use of instrumental variables increases statistical uncertainty, and we interpret the small observed differences in estimates as due to idiosyncratic statistical noise. Indeed, when the same specifications are compared, the estimates' confidence intervals overlap and typically include the estimates obtained with the other instrument.

[40]In Appendix Table A8, we reproduce Table 2 but add location-specific time trends as controls (i.e., we estimate a linear trend in time for each geographical unit). This is meant to account for potentially diverging electoral trends in different areas of the country, but it reduces the amount of available variation to estimate the main parameters of interest. Results for the DC are slightly larger than in our main specification and significant at the 1% level for all sets of controls, whereas the estimates for the PCI lose significance and become slightly positive. However, the difference in impact between DC and PCI is remarkably similar to our favorite specification.

[41]We define turnout as the number of citizens who cast a ballot – whether valid, blank, or invalid – divided by the

is negatively impacted by exposure to U.S. movies, in line with previous studies documenting a crowding-out of entertainment across different media (Falck et al., 2014; Gentzkow, 2006), but this margin does not drive our results. The decline is similar in PCI-leaning and DC-leaning areas, and an accounting decomposition of the eligible electorate shows that the gain for the DC and the loss for the PCI reflect a substitution of votes between the two blocs rather than differential turnout (Appendix Table A11, discussed in Appendix A).[42] Results on the vote share of other parties are close to zero in almost all cases, suggesting that exposure to U.S. movies before elections had effects only on the DC-PCI dimension. This, jointly with the historical context, provides additional supporting evidence that the results on the DC and the PCI vote shares are not spurious.

To better position these magnitudes within the media economics literature, we adapt the framework of DellaVigna and Gentzkow (2010) to compute an annualized persuasion rate. We estimate that an average voter's yearly diet of American cinema persuaded about 1.2% of uncommitted voters to shift their support to the DC in our OLS baseline, with IV estimates ranging from 1.1% to 2.0%. While this per-viewer rate is an order of magnitude smaller than the 8% to 12% rates typical of direct partisan news or political endorsements, the ubiquitous reach of Hollywood soft power allowed this subtle influence to compound into the substantial aggregate electoral shifts observed here. A detailed discussion of this methodology and the underlying calculations is provided in Appendix C.

Heterogeneous Effects. We next analyze whether the effects of exposure to U.S. movies before elections are heterogeneous across different dimensions. We split the sample according to a dimension of interest and separately re-apply our empirical strategy, at the cost of reduced sample sizes and, consequently, smaller F-statistics and less precise IV estimates. Appendix A.2 reports the results.

Splitting the sample by electoral leaning in the 1948 elections, we find that the increase in the

number of citizens eligible to vote (Section 3). This denominator differs from the one used for party vote shares, which are computed as votes for the party divided by the total number of valid votes.

[42] In Appendix Section A.3, we discuss our analysis of data collected by Costalli and Ruggeri (2019) on grassroots participation in PCI. We do not find significant impacts of exposure to U.S. movies.

DC vote share is driven by gains in areas that initially leaned toward the PCI (Appendix Table A12). Splitting it instead between Province Capitals and the Rest of the Province, the decrease in the PCI vote share is concentrated outside of the Province Capitals (Appendix Table A13). Since Province Capitals have much lower rates of agricultural employment, these estimates suggest that more agricultural areas experienced relatively stronger losses for the PCI. Splitting the sample by macro-region, instead, reveals no clear pattern, possibly due to the greater reduction in sample size.

6 Mechanisms and Content Analysis of Film Themes

The results in Section 5 establish that exposure to Hollywood movies increased support for the DC and decreased it for the PCI. In this section, we investigate the mechanisms through which film exposure may have influenced political preferences. We argue that cultural products such as movies reflect and transmit values, and that exposure to these products can reshape viewers' preferences and political attitudes. Election outcomes then reflect these changes, as voters are more likely to support the party that is most aligned with their resulting ideology and policy preferences.

We provide evidence for each step to support this mechanism. First, we show that Italian culture was substantially exposed to and influenced by American movies, while Soviet movies did not have a comparable influence.[43] Prominent American actors and "Hollywood" itself were heavily referenced in Italian cultural productions beginning after WWII, whereas references to Soviet actors, films, and cultural symbols remained much less widespread. We then show that American movies conveyed different values than Italian movies. This distinction is important because greater exposure to American movies necessarily implied lower relative exposure to domestic productions, which accounted for roughly 50% of movie theater revenues during the sample period. We analyze the plot summaries of American and Italian films using a state-of-the-art LLM, Gemini 3 Pro, following recent work on related topics (Jia and Strömberg, 2025). Relative to Italian movies, American films are more likely to highlight individualism and agency, meritocracy, and positively

[43] Section 3.2 showed that U.S. movies represented 40% of movie theater revenues in Italy during the studied period, while productions from all other countries combined accounted for only about 10% of revenues.

represent entrepreneurship and market dynamics, while being less likely to positively portray class conflict and redistribution or egalitarianism. Finally, after having established both the distinct value content of American movies and their broad diffusion within Italian culture, we show that the language used in American movies is more similar to that of conservative parliamentary speeches than the language used in Italian movies, which is comparatively more left-wing in its ideological content.

Finally, we complete the analysis of the proposed mechanism by examining the ITANES survey on Italians' social, political, and economic preferences. This analysis shows that people who express greater support for the U.S. than for the Soviet Union are also more likely to support the DC over the PCI. It also shows that exposure to American movies reduced support for greater independence from the U.S., both in general and in the domain of foreign policy, suggesting that exposure to U.S. movies shifted policy preferences toward positions more consistent with the political platform of the DC and away from that of the PCI.

In the rest of the section and in Appendix B, we provide full details on the methods used for the analysis and findings. Data sources are discussed in Section 3.

6.1 American Cultural Presence in Italy

We first document the diffusion of American popular culture into Italian society during our study period. Using Google Books Ngram data, we track the frequency of American cultural movie references in Italian-language publications from 1940 to 1980. The top panels of Figure 2 plot the frequency of references to prominent Hollywood stars – John Wayne, Marilyn Monroe, and Humphrey Bogart – in Italian books. All three names were virtually absent before the mid-1950s but increased sharply thereafter. John Wayne, the quintessential Western star embodying American individualism, exhibits the earliest and strongest growth. The term "Hollywood" itself increases nearly fivefold in Italian publications over this period, with growth accelerating markedly after 1955. This pattern confirms that American popular culture, and Hollywood in particular, achieved substantial penetration into Italian intellectual and cultural discourse during our study period.

Importantly, this cultural penetration was not shared by Soviet cultural figures. The two bottom panels of Figure 2 compare references to Hollywood and American cultural figures with their Soviet counterparts. The term “Hollywood” appears roughly eight times more frequently than “cinema sovietico” in Italian publications by 1980, despite the Cold War context that might have promoted Soviet cultural alternatives. The contrast in individual figures is even starker: John Wayne is mentioned approximately forty times more frequently than Sergei Eisenstein, the most internationally renowned Soviet filmmaker. This asymmetry suggests that exposure to American popular culture was not merely one of many foreign cultural influences, but a dominant cultural force.

6.2 Thematic Differences Between American and Italian Films

To examine whether American and Italian films differ on dimensions directly relevant to political attitudes, we analyze the value orientations embedded in film narratives. Following recent work using LLM-based content analysis of cultural texts (Jia and Strömberg, 2025), we annotate plot summaries from IMDb for each film in our sample using Gemini 3 Pro, coding value orientations along channels that map onto the individualism-collectivism divide central to Cold War political competition. We provide details on the methodology adopted in Appendix Section B.1.

Two films illustrate how these value orientations manifest in narrative content. *Harlow* (1965, American), depicting Jean Harlow’s rise in Hollywood, shows a protagonist who “starts in the movies as set dressing” but through persistence “finally graduates to featured roles” and “begins her climb to the top.” The film scores 7.3 on the Individualism Index (99.5th percentile), with maximum scores on both meritocracy (2.0) and agency (2.0). Narratives of individuals overcoming obstacles through talent, determination, and moral integrity exemplify the value orientations that distinguish American cinema.[44]

At the opposite extreme, Mario Monicelli’s Academy Award-nominated movie *I compagni* (1963, Italian) depicts textile factory workers organizing a strike in late 19th-century Piedmont.

[44] A similar rags-to-riches narrative of individual ambition and self-made success is portrayed in *Mister Cory* (1957, American), telling the story of “an ambitious Chicago slum kid with a knack for gambling” who “eventually rises to a high position in the world of professional gambling.” The film scores 6.2 on the Individualism Index (99th percentile), with high marks on meritocracy (1.8) and agency (1.8).

The plot centers on collective action: "impoverished employees labor fourteen hours a day" until "a former high school teacher turned labor organizer is dispatched to Turin to direct a strike." The film scores -8.9 on the Individualism Index (the sample minimum), with maximum scores on labor solidarity (2.0) and class conflict (2.0). The narrative emphasizes worker solidarity and collective struggle against exploitation over individual triumph – themes central to Italian left-wing political discourse.

Table 6 presents the value orientation scores by country of origin. American films score significantly higher on individualist dimensions: agency/individualism (+0.06, $p < 0.01$), meritocracy (+0.19, $p < 0.01$), and market legitimacy (+0.18, $p < 0.01$). Italian films, by contrast, show higher scores on redistribution/egalitarianism, though this difference is not statistically significant.[45] The pattern is consistent with American cinema promoting narratives of individual success and entrepreneurship, while Italian cinema – produced in a context of strong labor movements and a powerful communist party – gives relatively more weight to collective themes. These differences are robust to controlling for year fixed effects. In regression analysis reported in Table 7, American films score significantly higher on both the composite index (+0.041, $p < 0.01$) and a global 1–10 individualism scale assigned directly by the annotator (+0.13, $p < 0.01$).

As a robustness check, we also analyze thematic content using unsupervised topic models (LDA and DMR) applied to film dialogue from subtitles. This analysis confirms broad thematic differences, without directly measuring the value orientations central to our mechanism: American films feature more Westerns and urban dramas of individual ambition, while Italian films feature more domestic comedies and worker-related content. Full topic modeling methods and results are presented in Appendix B.3 and B.4.

[45] One channel shows an unexpected pattern: American films score marginally higher on "labor solidarity and unions" ($p < 0.10$). This reflects the nature of the films rather than contradicting our main finding. Both countries have very few films with any labor content (about 96% score zero). The American films receiving high scores are primarily historical epics depicting collective resistance in non-contemporary settings – *Spartacus* (1960, slave rebellion), *Mutiny on the Bounty* (1962, naval revolt), and *Modern Times* (1936, Chaplin's Depression-era satire). The annotator correctly identifies themes of collective action against authority, but these narratives frame solidarity as historical drama rather than contemporary political relevance. Italian films with high labor scores, by contrast, depict actual labor organizing: *I compagni* (1963), *La Califfa* (1970), and *La classe operaia va in paradiso* (1971) all feature union strikes and worker mobilization in contemporary industrial settings.

6.3 Political Alignment of Film Content

To assess how American and Italian film content align with political discourse, we construct an ideology measure using parliamentary speeches. The approach follows recent work applying text embeddings to measure political positions (Rheault and Cochrane, 2020): we train a classifier to learn what distinguishes speeches by conservative politicians from those by left-wing politicians.[46] We then use the classifier on movie dialogue to ask whether American or Italian films more closely resemble the "conservative" pattern. Details on the procedure are in Appendix B.2.

Table 8 presents the key result: American films score significantly higher on the ideology axis than Italian films. The coefficient of 0.086 ($p < 0.001$) indicates that, controlling for release year, American film dialogue is positioned closer to DC discourse than Italian film dialogue.

To contextualize this magnitude: the ideology score has a standard deviation of approximately 1.0 across all movie dialogue chunks, so the US–Italy gap of 0.086 represents about 9% of a standard deviation. While modest in absolute terms, this difference is highly statistically significant across 56,505 chunk-level observations.[47] For comparison, the DC–PCI gap in party manifestos is 0.55, so movies – which usually contain no explicit political content – show a US–Italy gap that is roughly 15% as large as the gap between the manifestos of Italy's two major opposing parties.

Concrete examples illuminate how the classifier identifies ideological valence even in entertainment dialogue that makes no explicit political claims. The American film *The Detective* (1968, Frank Sinatra) includes the line: *"I happen to feel that people should try to work out their own problems"* – a statement of individual self-reliance that the classifier assigns a 98.8% probability of coming from a conservative speaker, placing it in the top 0.1% of all movie dialogue. Italian films, by contrast, contain passages scoring at the opposite extreme. In the sixth highest-grossing movie of 1955, *Don Camillo e l'onorevole Peppone*, depicting the rivalry between a Catholic priest

[46] We classify as conservative the DC and all right-wing parties: Movimento Sociale Italiano, Partito Nazionale Monarchico, Partito Democratico Italiano di Unità Monarchica, Partito Monarchico Popolare. This classification is due to Democrazia Cristiana's fundamental Catholic roots and contraposition to the Communist and Socialist parties, even though it is generally considered a centrist party. In the Appendix, we show that the findings are actually stronger when using speeches from DC alone. As left-wing, we classify: Partito Comunista Italiano (PCI), Partito Socialista di Unità Proletaria, PCI-PSIUP, Partito Comunista Italiano-Indipendenti.

[47] See the footnote on dialogue chunks in Section 3.1 (Film Dialogues) for how chunks are constructed.

and Communist mayor, a character urges: *"Buy the people's newspaper... the workers' newspaper. Support the fight for peace"* – receiving only a 3.0% probability of conservative origin. Francesco Rosi's *Le mani sulla città* (Hands Over the City, 1963) includes dialogue framing the sale of public land as *"a moral issue... affecting the entire city"* – also scoring 3.0%.

These findings suggest a mechanism for how Hollywood movies may have influenced Italian voters. American films, even without explicit political messages, conveyed themes of individual ambition, self-reliance, and material success that better aligned with values promoted by the DC. Italian films, produced in a country with a powerful communist party and strong labor movement, more often depicted workers and collective struggles. Italians exposed to more American films encountered cultural content reinforcing the worldview associated with conservative politics, potentially shifting preferences toward the DC.

6.4 Survey Evidence on Pro-U.S. Attitudes and Political Preferences

In order to provide evidence complementing our analysis on electoral choices, we conduct an analysis using individual-level survey data from ITANES for 1968 and 1972. ITANES provides a snapshot of Italians' social, political, and economic preferences in the later part of the studied period. This will allow us to directly measure the alignment between the DC and the U.S. and to use a similar empirical design to look at the impact of exposure to American movies on policy preferences.

The model we estimate is the following:

$$y_{i,l,t} = \gamma_t + \alpha_l + \beta Cinema_{l,t-1} + c\mathbf{X_{i,l,t}} + \varepsilon_{i,l,t} \tag{3}$$

where $y_{i,l,t}$ represents an outcome for individual i in location l in year t; $Cinema_{l,t-1}$ measures exposure to cinema in the year prior to the survey in location l; $\mathbf{X_{i,l,t}}$ is a set of socio-economic controls, including sex, age, education, and income level; $\varepsilon_{i,l,t}$ is a random error term. α_l and γ_t are location and election year fixed effects, respectively. We cluster standard errors at the location level.

Table 9 presents the results from the survey data analysis. In columns (1)-(4), we show that a higher exposure to U.S. movies is significantly associated with increased preferences for alignment between Italy and the U.S. on foreign affairs. This suggests that U.S. movies changed policy preferences of their viewers, aligning them to American goals. We then focus on asking directly about favor for different entities. Columns (5)-(8) show that a higher exposure to cinema is positively associated with a preference for the U.S. over the USSR and for the DC over the PCI, although the estimates do not achieve statistical significance. Finally, we show that people who express more favor toward the U.S. than toward the Soviet Union tend to favor the DC over the PCI as well. Columns (9) and (10) show that individuals who prefer the DC over the PCI also tend to prefer the U.S. over the USSR.

Survey data provide further evidence suggesting that a greater exposure to cinema is associated with increased alignment with and favor toward the U.S. This supports our suggested hypothesis that U.S. movies, by exposing Italian viewers to values aligned with American objectives, changed their policy preferences and shifted votes away from the PCI and toward the U.S.-aligned party, the DC.

7 Conclusion

This paper examines whether exposure to Hollywood movies influenced electoral outcomes in Cold War Italy, a setting where cinema was the dominant form of mass entertainment and political competition between a U.S.-endorsed party and a Soviet-aligned communist party was unusually intense. Combining a novel panel dataset on local cinema availability and U.S. movie revenues with election returns from 1953 to 1972, and complementing fixed-effects estimates with an instrumental variable strategy, we present causal estimates of the electoral consequences of soft power.

We find that greater pre-election exposure to U.S. movies raised the vote share of the DC and lowered that of the PCI. The magnitudes are small per voter, corresponding to an annualized persuasion rate of roughly 1.2% (1.1% to 2.0% across IV estimates), but they are precisely estimated and stable across specifications, robustness checks, and survey-based outcomes. Effects on other parties and on turnout are negligible, indicating that the influence of American cinema operated along the ideological axis that defined Cold War politics in Italy.

We also provide evidence on the mechanism behind these results. American popular culture achieved deep penetration into Italian discourse, while Soviet cultural products did not. Using content analysis of plot summaries and a classifier trained on parliamentary speeches, we show that American films were systematically more individualist than Italian films and that their dialogue more closely resembled the language of conservative Italian political discourse. Survey responses indicate that Italians more exposed to cinema reported preferences more closely aligned with the U.S. on foreign policy. Taken together, these results suggest that the electoral effects of Hollywood operated through the gradual diffusion of values and worldviews consistent with the platform of the DC and at odds with that of the PCI.

Several questions remain open. Data limitations prevent us from identifying which film content moved voters, and the rise of targeted digital media raises the possibility that individual susceptibility to foreign influence is now more heterogeneous than in the cinema era. The decline in American cultural exports documented by Fan et al. (2024) points to a further avenue of inquiry. To the extent that exposure to U.S. media shapes geopolitical alignment, its retreat may contribute to political divergence between the U.S. and other major powers. Whereas Kleinman et al. (2024) trace political alignment to the economic interdependence created by trade, our findings point to a complementary cultural channel, so that declining U.S. cultural exports could weaken alignment even where trade ties remain intact.

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Figures and Tables

Table 1: Summary Statistics

	Mean	Std. Dev	Min.	Max.	N
Cinema variables					
Movie theaters per 10,000 inhabitants	2.04	0.84	0.32	7.70	3738
Number of movie theaters	56.02	59.89	1.00	461.00	3738
Number of movie theaters in 1948	41.59	45.01	1.00	307.00	178
Tickets sold per capita	15.47	8.01	0.29	42.81	3738
Movies					
% revenues U.S. movies in Italy	39.65	12.89	27.33	64.14	21
% revenues Italian movies in Italy	50.06	12.23	23.03	65.15	21
% revenues Sovietic movies in Italy	0.21	0.20	0.00	0.64	21
% revenues French movies in Italy	2.55	1.57	0.44	7.11	21
% revenues German movies in Italy	1.35	0.98	0.09	4.30	21
% revenues U.S. movies in France	30.72	4.08	24.53	40.15	21
% revenues U.S. movies in the Netherlands	46.77	8.21	39.71	66.77	21
Electoral variables					
DC Share	37.71	9.32	10.46	68.10	890
PCI Share	25.31	10.63	2.32	61.14	890
Turnout	93.77	4.43	75.07	99.94	890
Census variables					
% employed in the agricultural sector	26.82	20.64	0.30	80.40	534
% employed in the industrial sector	36.41	11.28	10.66	75.86	534
% employed in the commercial sector	12.98	5.13	3.68	29.40	534
% employed in the services sector	23.79	12.45	4.65	57.30	534
Employment rate	46.14	5.74	32.70	64.16	534
Labor force participation	48.38	5.50	35.80	66.82	534
% with a college degree	7.06	4.13	1.39	19.10	534

Notes: Electoral variables refer to the national elections held in 1953, 1958, 1963, 1968, and 1972. Census variables are obtained from various Italian Decennial Censuses (1951, 1961, and 1971). Variables are averaged over the period 1950-1973.

Table 2: Effects of U.S. movies on electoral choices: OLS specification

	DC Share				PCI Share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% DC	% DC	% DC	% DC	% PCI	% PCI	% PCI	% PCI
Exposure	0.0120***	0.00856***	0.00757**	0.00751***	-0.00994***	-0.00561**	-0.00448**	-0.00447**
	(0.00326)	(0.00290)	(0.00294)	(0.00286)	(0.00231)	(0.00226)	(0.00226)	(0.00226)
Sectorial employment		✓	✓	✓		✓	✓	✓
Labor participation		✓	✓	✓		✓	✓	✓
Population			✓	✓			✓	✓
Education				✓				✓
Mean Dep. Variable	0.377				0.253			
SD Dep. Variable	0.0932				0.106			
Within R-sq	0.0287	0.0880	0.0927	0.122	0.0308	0.120	0.130	0.134
Locations	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890

Notes: OLS estimates from equation 1 in all columns. All specifications include election year and location fixed effects. The dependent variable is the vote share for the DC in columns (1)-(4) and for the PCI in columns (5)-(8). The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. Sectoral employment controls include controls for the share of people employed in the agricultural, industrial, commercial, and service sectors; labor participation controls include employment rate and labor force participation; population is the population logarithm; education indicates the share of people with a college degree. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table 3: Effects of U.S. movies on electoral choices: IV specification

	1^{st} Stage	DC Share				PCI Share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Exposure	% DC	% DC	% DC	% DC	% PCI	% PCI	% PCI	% PCI
Instrument	1.158***								
	(0.159)								
Exposure		0.0380***	0.0284***	0.0279***	0.0241***	-0.0357***	-0.0257***	-0.0248***	-0.0238***
		(0.00867)	(0.00894)	(0.00921)	(0.00859)	(0.00715)	(0.00804)	(0.00883)	(0.00876)
Sectorial employment			✓	✓	✓		✓	✓	✓
Labor participation			✓	✓	✓		✓	✓	✓
Population				✓	✓			✓	✓
Education					✓				✓
F-stat	53.22	53.22	32.38	28.61	29.00	53.22	32.38	28.61	29.00
Locations	178	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890	890

Notes: IV estimates from equation 2 in columns (2)-(9). All specifications include election year and location fixed effects. The dependent variable is the vote share for the DC in columns (2)-(5) and for the PCI in columns (6)-(9). The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. The instrument is the interaction term between local exposure to cinema in 1948 and the success of U.S. movies in France as explained in Section 4, and it is standardized. Sectoral employment controls include controls for the share of people employed in the agricultural, industrial, commercial, and service sectors; labor participation controls include employment rate and labor force participation; population is the population logarithm; education indicates the share of people with a college degree. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table 4: Robustness to alternative specifications

A) Dependent variable: DC Share

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Exposure	0.00751***	0.0246***	0.0241***	0.0355***	0.0358***	0.0512***	0.00820**	0.0141	0.00725	0.00712	0.0132**	0.0347
	(0.00286)	(0.00602)	(0.00859)	(0.0111)	(0.00988)	(0.0118)	(0.00397)	(0.00877)	(0.00592)	(0.0355)	(0.00522)	(0.0211)
Contemporaneous Exposure		✓		✓		✓		✓		✓		✓
Exposure measure	Cinema	Cinema	Cinema	Cinema	Cinema	Cinema	Tickets	Tickets	Tickets	Tickets	Tickets	Tickets
Method	OLS	OLS	IV	IV	IV	IV	OLS	OLS	IV	IV	IV	IV
Instrument Country			FR	FR	NL	NL			FR	FR	NL	NL
Locations	178	178	178	178	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890	890	890	890	890

B) Dependent variable: PCI Share

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Exposure	-0.00447**	-0.00485	-0.0238***	-0.0333**	-0.00937	-0.0116	-0.000793	-0.000319	-0.0111**	-0.0704**	-0.000813	-0.000615
	(0.00226)	(0.00506)	(0.00876)	(0.0131)	(0.00755)	(0.0106)	(0.00278)	(0.00672)	(0.00491)	(0.0310)	(0.00366)	(0.0137)
Contemporaneous Exposure		✓		✓		✓		✓		✓		✓
Exposure measure	Cinema	Cinema	Cinema	Cinema	Cinema	Cinema	Tickets	Tickets	Tickets	Tickets	Tickets	Tickets
Method	OLS	OLS	IV	IV	IV	IV	OLS	OLS	IV	IV	IV	IV
Instrument Country			FR	FR	NL	NL			FR	FR	NL	NL
Locations	178	178	178	178	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890	890	890	890	890

Notes: the Table reports results from various regressions aimed at testing the robustness of our results. All specifications include election year and location fixed effects. All regressions include controls for sectoral employment, labor participation, population, and education. Panel A shows results using DC Share as dependent variable while in Panel B the dependent variable is the vote share for the PCI. In Column (1) and (3) we report for convenience the same regression as Tables 2 and 3 respectively. The row Contemporaneous Exposure indicates whether we use the uninteracted contemporaneous local exposure to cinema. The row Exposure measure indicates which local exposure to cinema we use (cinema or tickets per capita). The row Method indicates whether we are estimating equation with OLS or IV. The row Instrument Country indicates which instrument we use (France or the Netherlands). Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table 5: Effects of U.S. movies on turnout and other parties

	(1) Turnout	(2) MSI	(3) PLI	(4) PRI	(5) PSI	(6) Monarchists	(7) Other parties	(8) DC	(9) PCI
Exposure	-0.00426**	-0.00375	-0.00152	0.000946	0.00459	-0.00559*	0.00114	0.00751***	-0.00447**
	(0.00189)	(0.00235)	(0.00183)	(0.00133)	(0.00350)	(0.00329)	(0.00168)	(0.00286)	(0.00226)
Sectorial employment	✓	✓	✓	✓	✓	✓	✓	✓	✓
Labor participation	✓	✓	✓	✓	✓	✓	✓	✓	✓
Population	✓	✓	✓	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mean Dep. Variable	0.938	0.0618	0.0444	0.0212	0.150	0.0361	0.0279	0.377	0.253
SD Dep. Variable	0.0443	0.0420	0.0315	0.0289	0.0500	0.0471	0.0614	0.0932	0.106
Within R-sq	0.250	0.0889	0.183	0.112	0.0750	0.207	0.149	0.122	0.134
Locations	178	178	178	178	178	178	178	178	178
N	890	890	890	890	890	704	890	890	890

Notes: OLS estimates from equation 1 in all columns. All specifications include election year and location fixed effects. The dependent variable is the vote share of various parties as indicated in the heading of the column. The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. Sectoral employment controls include controls for the share of people employed in the agricultural, industrial, commercial, and service sectors; labor participation controls include employment rate and labor force participation; population is the population logarithm; education indicates the share of people with a college degree. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table 6: Value Orientations in Film Plot Summaries by Country of Origin

	US	Italy	US–Italy Δ
Individualist channels			
Agency: individual vs. collective	+0.748	+0.686	+0.062***
Meritocracy/self-made success	+0.501	+0.313	+0.188***
Market/entrepreneurship legitimacy	+0.174	−0.003	+0.177***
Consumerism/material aspiration	+0.382	+0.160	+0.222
Collectivist channels			
Labor solidarity & unions	+0.586	+0.439	+0.147*
Redistribution/egalitarianism	+0.473	+0.764	−0.291
Class-conflict salience	−0.468	−0.395	−0.072

Notes: Each cell reports the mean framing score = direction ($-1/0/+1$) $\times$ salience (0–2) $\times$ confidence (0–1), aggregated by country of origin. Positive values indicate pro-value framing; negative values indicate anti-value framing; near-zero values indicate low salience. Films are weighted by inverse box office rank within season (normalized within season). The US–Italy Δ column reports the difference between American and Italian films; significance stars come from an unweighted Welch t-test ($^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$). Sample: $N = 1{,}647$; US$=750$; Italy$=897$.

Table 7: American Films Score Higher on Both Value Orientation Measures

	(1) Individualism Index	(2) Individualist–Collectivist Scale
US origin (vs. Italy)	0.041***	0.128***
	(0.015)	(0.049)
Year FE	Yes	Yes
Observations	1,594	1,461
R^2	0.034	0.036

Notes: Column (1): Dependent variable is the Individualism Index, computed as the sum of individualist channel scores (agency, meritocracy, market legitimacy, consumerism) minus collectivist channel scores (labor solidarity, redistribution, class conflict). Each channel score equals direction $\times$ salience $\times$ confidence. Column (2): Dependent variable is the global Individualist–Collectivist scale (1–10, higher values are more individualist), assigned directly by the LLM based on overall plot content. Both measures derived from LLM annotation of plot summaries. Unit of observation is a film. Sample restricted to U.S. and Italian productions. Heteroskedasticity-robust standard errors in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 8: American Films Score Higher on Political Ideology Axis

	(1)	(2)
US (vs. Italy)	0.086***	0.086***
	(0.008)	(0.008)
Year FE	No	Yes
Observations	56,505	56,505
R^2	0.002	0.006

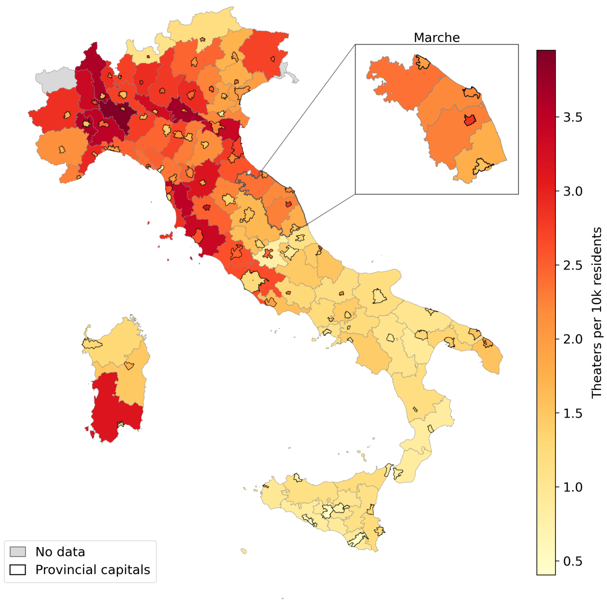
Notes: Each column regresses the chunk-level ideology score on U.S., a dummy equal to one for dialogue chunks from American films and zero for Italian films; the coefficient is the average US–Italy difference in ideology score. The ideology score is the linear predictor from a logistic classifier trained on Italian parliamentary speeches (1948–1974), embedded with a multilingual sentence encoder, to distinguish conservative parties (DC, MSI, monarchist) from left-wing parties (PCI, PSIUP); higher values denote dialogue closer to conservative parliamentary discourse (see Appendix B.2). The unit of observation is a subtitle dialogue chunk ($\approx$10–15 lines); $N = 56{,}505$ across U.S. and Italian films. Column (2) adds release-year fixed effects. Heteroskedasticity-robust standard errors in parentheses. $^{***}p < 0.01$.

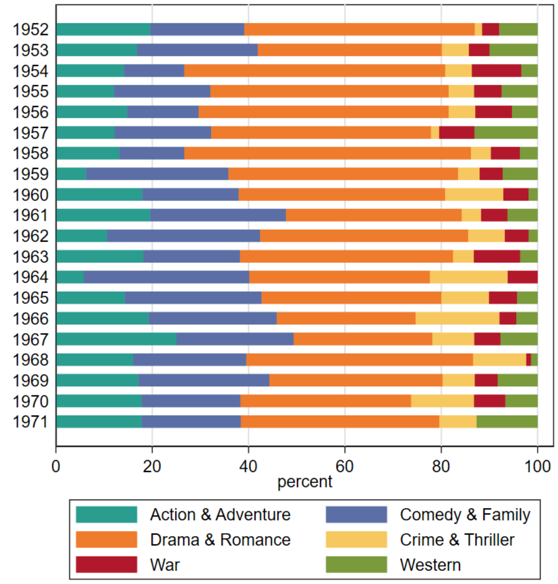
Table 9: Survey data results

	USA				Preference Gaps					
	(1) Independence	(2) Independence	(3) Foreign Policy	(4) Foreign Policy	(5) US-URSS	(6) US-URSS	(7) DC-PCI	(8) DC-PCI	(9) US-URSS	(10) US-URSS
Cinema Exposure	0.338***	0.336***	0.249**	0.235*	0.179	0.172	0.119	0.111		
	(0.126)	(0.125)	(0.119)	(0.119)	(0.126)	(0.124)	(0.184)	(0.174)		
DC-PCI									0.609***	0.589***
									(0.0241)	(0.0254)
Age		0.00449***		0.00532***		0.00759***		0.00972***		0.00188
		(0.000752)		(0.000777)		(0.00126)		(0.00112)		(0.00123)
Female		0.0237		0.0666***		0.287***		0.309***		0.126***
		(0.0208)		(0.0209)		(0.0361)		(0.0431)		(0.0298)
High School		0.0464		0.143***		0.250***		0.188***		0.117**
		(0.0321)		(0.0321)		(0.0530)		(0.0478)		(0.0524)
University		0.000148		0.193***		0.238***		0.231***		0.0880
		(0.0439)		(0.0459)		(0.0845)		(0.0612)		(0.0800)
Location FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Survey FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Income FE		✓		✓		✓		✓		✓
Mean Dep. Variable	0.431		0.403		0		0		0	
SD Dep. Variable	0.495		0.491		1		1		1	
R-sq	0.149	0.167	0.122	0.163	0.109	0.151	0.139	0.188	0.424	0.432
Locations	140	140	140	140	143	143	141	141	141	141
N	2174	2174	2154	2154	3127	3127	2871	2871	2804	2804

Notes: The table illustrates the effect of exposure to cinema on individual political preferences, using survey data from the 1968 and 1972 waves of the Italian National Election Study (ITANES). The dependent variable in columns (1) and (2), Independence, is a dummy variable which is equal to one when the respondent answers that independence of Italy from the U.S. is either of little importance or not at all important; in columns (3) and (4), Foreign Policy, is a dummy equal to one when the respondent disagrees with the statement that *"the foreign policy of Italy ought to be completely independent of the United States"*; in columns (5), (6), (9), and (10), US-USSR, is a standardized variable measuring the difference in how much the individual expresses appreciation for the U.S. or the USSR (higher values correspond to a preference for the U.S.); in columns (7) and (8), DC-PCI, is defined analogously to the previous variable but the groups of appreciation are now the DC and the PCI. Income FE refer to a fixed effect included for each of the eight brackets of income present in the survey. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

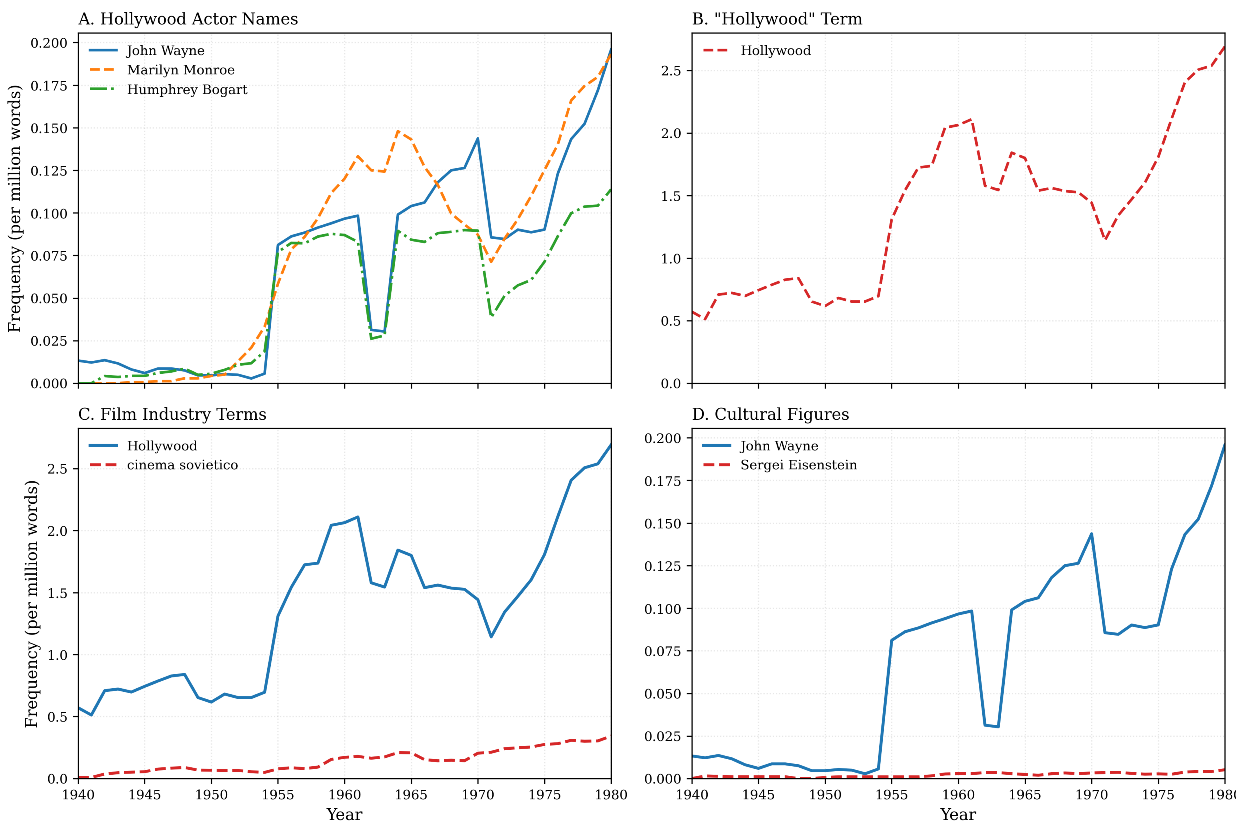
Figure 1: Distribution of Movie Theaters in 1948 and Genre Composition over Time





Notes: The left panel shows the number of movie theaters per 10,000 inhabitants in 1948 in Italy for each of our units of analysis. For each province in the sample, the paper defines two units of observation: (i) the Province Capital (i.e., the main municipality), highlighted by a dark border, and (ii) the Rest of the Province excluding the Province Capital, delimited by a light gray border. Thus, each province contributes two observations to the analysis: the Province Capital and the Rest of the Province. We show an enlarged version of the Marche region for easier inspection of the difference between the two types of location. The right panel displays the composition of U.S. movies' revenues per year of production by movies' genre. Genre is obtained using data from the publication *Variety*.

Figure 2: American and Soviet Cultural References in Italian Books, 1940–1980



Notes: Data from Google Books Ngram Viewer (Italian corpus, 2019 version). Y-axis shows frequency of term per million words, smoothed with 3-year moving average. Panel A shows prominent Hollywood actor names; Panel B shows the term "Hollywood" on a separate scale. Panel C compares film industry terms; Panel D compares prominent cultural figures (American actor vs. Soviet director).

Online Appendix to Electoral Effects of U.S. Soft Power: Evidence from Hollywood Movies in Cold War Italy

A Additional Results and Robustness Checks

This appendix presents in detail the robustness checks and heterogeneity analyses summarized in Section 5.

A.1 Robustness Checks

To understand whether the prolonged exposure might affect voters and bias our short-term estimates, we explore whether exposure two years prior to the election is related to the outcomes of interest and study its ability to explain the reported short-term results. Columns (2) and (7) of Appendix Table A9 show that exposure beyond one year from the election is not strongly related to voting outcomes, while columns (1) and (6) show that it does not meaningfully affect the magnitude of the short-run coefficient when included alongside shorter-term exposure. We conclude that exposure to Hollywood movies only affected voters in the short term, with little to no persistence of effects.

In Appendix Table A9, we also look at the effect of exposure in the year of the election. As discussed in Section 4, we use the year before elections to measure exposure in our main analysis since all elections in our sample are held between March and May, while box office data refer to calendar years. This means that exposure measured in the election year includes mostly months of post-election exposure. In columns (4) and (9), we show that exposure measured in the election year is not significantly related to voting outcomes: estimated coefficients have the same sign as the parameters obtained with our main exposure measure, but their magnitude is approximately halved. Columns (3) and (8) show that it does not meaningfully affect the magnitude of the main coefficient of interest when they are both included in the regressor. The decrease in significance of the parameter for PCI is only due to an increase in statistical uncertainty (probably caused by the increase in parameters estimated), and not by a decrease in the estimated magnitude of the impact; both magnitude and significance for the impact on the vote share of DC are largely unaffected.

Appendix Table A7 reports the baseline estimates with standard errors that account for spatial correlation using the procedure developed in Conley (1999). Specifically, columns (1)-(4) and (5)-(8) replicate columns (4) and (8) from Table 2 respectively, using different distance cutoffs. Standard errors are similar in size, and overall significance patterns are unaffected.

We conduct a falsification test by randomly permuting the values of the two separate components of our explanatory variable, local access to cinema and the success of U.S. movies, 10,000 times, and create a surrogate dataset for each step. We then re-estimate our favorite specification for each dataset and store the estimates. Appendix Figure A2 displays the distribution of the estimated coefficients for the DC (top panel) and the PCI vote shares (bottom panel). The distribution of the estimated coefficients is centered around zero, and our estimates from Table 2 (Panel A, columns 4 and 8), clearly lie in the extreme parts of the respective distributions.

We then examine more closely the reduction in turnout documented in Table 5. A one standard deviation increase in exposure lowers turnout by about 0.4 percentage points. This effect is comparable in absolute size to the effect on the PCI vote share, but modest relative to a baseline turnout of roughly 94%, and its magnitude is consistent with the entertainment crowding-out channel documented in other studies (Falck et al., 2014; Gentzkow, 2006). Since a selective decrease in turnout could potentially change the relative shares of the DC and the PCI, we first look for direct evidence of such selection, and then re-estimate our main model expressing the outcomes as shares of eligible voters rather than of valid votes.

Panel A of Appendix Table A11 shows that the impact on turnout is not concentrated in left-leaning areas. Splitting locations by their 1948 political leaning, the turnout decline is similar in PCI-leaning and DC-leaning areas, and the interaction between exposure and an indicator for DC-leaning areas is small and statistically insignificant. The reduction in turnout is therefore broad-based, as one would expect from a crowding-out mechanism, rather than a selective demobilization of PCI supporters. Even in the extreme case in which all of the exposure-induced abstention were attributed to would-be PCI voters – a case clearly rejected by the estimates in Panel A – differential turnout could account for at most about three-quarters of the estimated PCI decline and less than a quarter of the DC gain.

Panel B of Appendix Table A11 re-expresses the outcomes as shares of the eligible electorate, partitioning the electorate into the DC, the PCI, all other parties, and abstention. Consistent with our main analysis, the DC bloc grows and the PCI bloc shrinks by similar magnitudes, while the increase in abstention is drawn from the residual category rather than from the PCI. The main results are then caused by a substitution of votes between the two blocs, and are not an artifact of differential turnout.

A.2 Heterogeneous Effects

In Appendix Table A12, we split our sample between "DC areas" and "PCI areas" according to the votes received by the two parties in the 1948 elections.[48] Results suggest that the increase in the DC vote share is driven by gains in PCI areas.

In Appendix Table A13, we present the results where we split the sample between Province Capitals in Panel (a) and Rest of the Province in Panel (b). The results are consistent with the main findings and show that the decrease in votes for the PCI is mainly driven by losses outside of the Province Capitals. It is worth noting that our geographical units of analysis generally include both urban and rural areas. However, as shown in Appendix Table A6 Province Capitals have much lower rates of agricultural employment, and these estimates can be interpreted as suggesting that more agricultural areas experienced relatively stronger losses for the PCI.

We also split the sample along the geographical dimension in Appendix Table A14, but do not observe any clear and informative pattern, possibly due to a greater reduction in sample size.

[48] We call "DC areas" the areas where the ratio of DC votes over the sum of DC votes and PCI votes is above the sample median in 1948, and "PCI areas" the other half.

A.3 Effects on Grassroots Participation in PCI

To further investigate a potential mechanism driving our results, we use data measuring the organizational strength and the grassroots participation in the PCI from Costalli and Ruggeri (2019). This source, obtained from internal reports of the PCI, provides the number of party members per year in every Italian province and the number of local PCI offices in a given province from 1951 to 1968. Since information is available only at the province level, the cross-sectional dimension of our panel is halved.[49] On the other hand, the fact that data are available at a higher frequency (yearly) with respect to national elections (every five years) offers more time variation. No comparable data exist for the DC, to the best of our knowledge.

Appendix Table A10 presents results on local party organization – columns (1) to (4) – and local membership in the PCI – columns (5) to (8). In both cases, the effects are close to zero, suggesting that the loss in votes for the PCI arising from a higher exposure to U.S. movies was not mediated through organizational disadvantages or lower grassroots political participation. However, the estimates are imprecise, making the exercise mostly inconclusive.

B Content Analysis Methodology

B.1 Plot Summary Data and Annotation.

We obtain plot summaries from IMDb for films in our box office sample, selecting the longest available summary for each title.[50] We use Gemini 3 Pro to annotate each plot along seven value-orientation channels, following recent work on LLM-based content analysis of cultural texts (Jia and Strömberg, 2025).

For each channel, the model assigns: (i) a salience score (0–2) indicating how prominently the theme features (0 = absent, 1 = present, 2 = central); (ii) a direction (-1, 0, or $+1$) indicating whether the narrative frames the value negatively, neutrally, or positively; and (iii) a confidence

[49] For this part of the analysis, we aggregate local exposure to cinema at the province level using information from SIAE from the main city and the rest of the province to make the data comparable.

[50] The longest summary usually contains the most narrative detail, which helps the annotation.

score (0–1). The channel score is direction $\times$ salience $\times$ confidence. The seven channels divide into individualist-oriented and collectivist-oriented themes.

Individualist channels:

- **Agency/individualism**: Whether protagonists shape their own destiny through personal choices vs. being swept along by social forces or group decisions.
- **Meritocracy/self-made success**: Narratives where hard work, talent, or determination lead to advancement regardless of birth or connections.
- **Market/entrepreneurship legitimacy**: Positive portrayals of business, profit-seeking, or commercial success.
- **Consumerism/material aspiration**: Desire for goods, wealth, or lifestyle advancement as legitimate goals.

Collectivist channels:

- **Labor solidarity and unions**: Depictions of worker organizing, collective bargaining, or union activity.
- **Redistribution/egalitarianism**: Themes of sharing wealth, reducing inequality, or critiquing economic disparities.
- **Class conflict salience**: Narratives structured around tension between social classes (workers vs. owners, poor vs. rich).

For instance, a protagonist who "works his way through the Police Academy" scores high on meritocracy, while a plot where "a labor organizer directs a strike" scores high on labor solidarity.

Aggregate Measures. The *Individualism Index* sums individualist channel scores and subtracts collectivist channel scores, with theoretical range from -14 to $+14$ (the observed range is -8.9 to $+8.8$). In addition to the feature-specific coding, the model assigns a global *Individualist–Collectivist scale* (1–10, higher values being more individualist) based on its holistic assessment of the overall narrative – this single rating captures the annotator's direct judgment of where the film falls on the individualism–collectivism spectrum, without requiring the researcher to specify weights across channels.

To illustrate the scale, on the individualist end the model assigns a 10 to American *Cool Hand Luke* (1967), the story of a defiant lone convict on a chain gang, and to Italian *Per un pugno di dollari* (1964), Sergio Leone's archetypal lone-gunslinger spaghetti Western (rank 2 of the 1964–65 season). At the opposite end, the lowest-scoring Italian films include Mario Monicelli's *I compagni* (1963) – already discussed in Section 6.2 – and *Il Vecchio Testamento* (1963), a peplum centered on the Maccabean revolt against the Seleucid Empire. The bottom of the Italian distribution is occupied by a labor-organizing narrative and a religious epic of collective resistance against foreign rule, echoing the channel-level results in which departures from individualist framings tend toward collective struggle.

Model and Validation. Annotations were generated using Gemini 3 Pro. The prompt instructs the model to use only information in the plot summary, without relying on outside knowledge of the film. Evidence excerpts are required to be verbatim substrings from the plot text. For sample construction and coverage rates, see Section 3.1.

Regression Results. Table 7 reports regressions of our two value-orientation measures on a U.S.-origin dummy, with year fixed effects. Both columns confirm the pattern in Table 6: American films score significantly higher than Italian films on individualism. The Individualism Index (column 1) is 0.041 points higher for U.S. films ($p < 0.01$), and the LLM-assigned 1–10 Individualist–Collectivist scale (column 2) is 0.128 points higher ($p < 0.01$). These gaps correspond to about 0.02 and 0.07 of a standard deviation, reflecting that the country-of-origin signal is concentrated on a few of the seven channels (agency, meritocracy, market legitimacy) rather than spread uniformly across them. The fact that the gap shows up on both the bottom-up composite index and the holistic LLM rating – two outcomes constructed in very different ways – suggests the difference is a feature of overall narrative content rather than an artifact of how we weight specific channels.

The estimates are robust to year fixed effects. Consistent with the contrast between *Harlow*/*Mister Cory* and *I compagni* discussed in Section 6.2, the regressions formalize what the channel-by-channel comparison already suggests: American films are systematically more individualist than Italian films across both aggregate measures.

B.2 Political Alignment Classifier

Procedure. Our procedure follows four steps. First, we split parliamentary speeches into sentence-level segments and label each speaker's party as "conservative" ($y_i = 1$) or "left-wing" ($y_i = 0$). The conservative block comprises Democrazia Cristiana, Movimento Sociale Italiano, Partito Nazionale Monarchico, Partito Democratico Italiano di Unità Monarchica, and Partito Monarchico Popolare; the left-wing block comprises Partito Comunista Italiano, Partito Socialista di Unità Proletaria, PCI-PSIUP, and Partito Comunista Italiano-Indipendenti. We exclude centrist parties from this exercise.[51] Second, we represent each segment as a dense 1,024-dimensional embedding $\mathbf{x}_i \in \mathbb{R}^{1024}$ using the multilingual sentence encoder `multilingual-e5-large`, which maps semantically similar text snippets to nearby points in space regardless of language.[52] Third, we fit a logistic regression classifier on these embeddings to predict party family,

$$\Pr(y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\alpha + \mathbf{x}_i \cdot \beta))},$$

where α is a scalar intercept and β is the coefficient vector defining the separating hyperplane in the embedding space. We define the logit ideology score $\hat{s}_i$ as the estimated linear predictor,

$$\hat{s}_i = \hat{\alpha} + \mathbf{x}_i \cdot \hat{\beta},$$

[51] Democrazia Cristiana is generally considered a centrist party, but its foundational Catholic roots and contraposition to the Communist and Socialist parties support its classification with other conservative parties.

[52] An important feature of this approach is that parliamentary speeches and party manifestos are in Italian, while movie subtitles are in English. The multilingual encoder maps both languages into a shared semantic space, enabling cross-lingual comparison. If we had embedded Italian-language dialogue for Italian films and English dialogue for American films, one might worry that the encoder places Italian texts closer to Italian political discourse simply due to shared language. By using English subtitles for *all* films, any US–Italy difference we detect reflects thematic content, not language proximity. See Section 3.1 for the construction of the subtitle sample.

so that positive scores indicate proximity to conservative speech patterns and negative scores indicate proximity to left-wing patterns. Fourth, we use the same embedding model to generate the $\mathbf{x}_i \in \mathbb{R}^{1024}$ covariates from party manifesto or movie dialogue chunks, and then apply the trained classifier to obtain a predicted probability and score of conservative origin.

Validation with Party Manifestos. Before applying this measure to films, we validate it on party manifestos, documents unseen by the classifier during training. If the method captures meaningful political content, DC manifestos should score as more "conservative" than PCI manifestos. Table A1 confirms this: DC manifesto sentences score 0.55 points higher ($p < 0.001$), correctly distinguishing the two parties.

Table A1: Validation – Ideology Measure Distinguishes DC from PCI Manifestos

	(1)	(2)
DC (vs. PCI)	0.429^{**}	0.550^{***}
	(0.169)	(0.145)
Year FE	No	Yes
Observations	2,833	2,833
R^2	0.031	0.071

Notes: Dependent variable is ideology score (higher = more right-wing). Unit of observation is a manifesto sentence. Classifier trained on parliamentary speeches (1948–1974) to distinguish right-wing from left-wing parties. Standard errors clustered by manifesto in parentheses. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

What does the measure capture? Passages scoring highest (most "conservative") discuss institutional procedures, economic stability, and state capacity – e.g., *"La stabilità del sistema economico è garantita"* ("The stability of the economic system is guaranteed," DC 1968). Passages scoring lowest invoke class struggle and anti-imperialism – e.g., *"L'aggressione dell'imperialismo americano contro l'eroico popolo del Vietnam"* ("American imperialism's aggression against the heroic Vietnamese people," PCI 1968) or references to *"le masse lavoratrici"* ("the working masses").

Robustness: DC vs. PCI Only. In an alternative specification, we train the classifier on DC versus PCI speeches only, excluding all other parties. The US–Italy difference in film ideology scores increases to 0.211 ($p < 0.001$), suggesting that our main specification – which pools right-wing parties with the DC and the broader left-wing block – is conservative.

B.3 Topic Modeling (LDA)

As a complementary analysis of film content, we characterize thematic differences using Latent Dirichlet Allocation (LDA; Blei et al., 2003), an unsupervised topic model applied to film dialogue from subtitles. LDA has been increasingly used for large-scale text analysis in economics (see Ash and Hansen, 2023, for a survey), with applications to cultural texts (Michalopoulos and Xue, 2021) and film content (Michalopoulos and Rauh, 2024).

LDA is a generative probabilistic model that represents each document as a mixture over K topics, where each topic is a distribution over words. Given a corpus, LDA infers: (i) the topic distribution θ_d for each document d; and (ii) the word distribution ϕ_k for each topic k. We estimate the model using variational inference on English subtitle text from 1,109 films.[53]

We preprocess subtitle text by lowercasing, removing stopwords, and lemmatizing. To ensure topics reflect substantive themes rather than film-specific proper nouns, we mask named entities (e.g., persons, places, nationalities) prior to estimation, consistent with standard topic-modeling practice (Gentzkow et al., 2019; Mueller and Rauh, 2018; Cormun and Ristolainen, 2024). We estimate models with varying numbers of topics ($K \in \{10, 15, 20, 25\}$) and select $K = 10$ based on topic coherence scores and interpretability; results are qualitatively similar across specifications.

Results. American films are dominated by Westerns – the frontier mythology of rugged individualism and self-made fortunes. This topic alone accounts for over 35% of content in American films but only 4% in Italian productions. American films also feature prominently urban dramas depicting hustlers, corporate climbers, and individual ambition. Italian films more frequently feature domestic comedies and genre productions. Table A2 presents the full topic results.

[53] See Section 3.1 for details on the subtitle sample data construction.

Table A2: Full Topic List from LDA Model ($K = 10$)

#	Label	Top Words	IT	US	IT–US
0	Heists & crime	thief, knife, bank, hole, robbery, steal, million, principal	0.024	0.007	+0.02
1	Courtroom drama	court, murder, witness, guilty, trial, evidence, lawyer, crime	0.120	0.117	+0.00
2	Historical epics	king, slave, soldiers, army, gods, sire, master, power	0.113	0.077	+0.04
3	Period/religious	thee, thy, thou, church, lord, devil, soul, priest	0.065	0.069	−0.00
4	Westerns & frontier	ain't, horses, gold, train, folks, goin', guns, fellow	0.036	0.351	−0.32
5	Crime/thriller	bank, cop, hotel, madame, clock, million, cops, inspector	0.038	0.048	−0.01
6	Urban relationships	apartment, picture, kidding, ball, cat, key, lunch, sex	0.042	0.107	−0.07
7	Domestic comedy	professor, mom, dad, music, letter, allow, hotel, sing	0.334	0.034	+0.30
8	Naval/military	ship, captain, officer, general, boat, army, enemy, command	0.041	0.155	−0.11
9	Genre adventure	gold, dollars, million, bank, boss, thousand, sheriff	0.187	0.034	+0.15

Notes: Topics from LDA with $K = 10$ on masked subtitle text. "IT" and "US" columns show mean topic share for Italian-origin and U.S.-origin films respectively. "IT–US" is the difference in prevalence. Top words selected by highest probability within each topic's word distribution (ϕ_k). Labels assigned by authors based on top words and representative documents.

B.4 Dirichlet Multinomial Regression (DMR)

As an alternative approach to topic modeling, we estimate a Dirichlet Multinomial Regression (DMR) model on film dialogue. DMR differs from standard LDA in how it handles document-level covariates.

Methodological Distinction. In standard LDA, all documents are treated as exchangeable, as they share a single Dirichlet prior over topics. To compare topic prevalence between American and Italian films, one estimates the model and then computes *sample averages* of topic proportions by country. The difference is a post-hoc statistic.

DMR, differently from LDA, parametrizes the Dirichlet prior itself as a function of covariates:

$$\theta_d \sim \text{Dirichlet}(\exp(X_d \lambda))$$

where X_d contains indicators for country of origin and release year, and λ are learned coefficients. This means the *expected difference* in topic prevalence between American and Italian films is a model parameter estimated directly within the generative process, not a sample statistic computed after the fact. DMR thus controls for year effects in the data-generating process itself in addition to US/IT topic prevalence, avoiding potential confounds from temporal trends in filmmaking.

Results. The DMR results confirm our main findings. American films show significantly higher prevalence of Western/frontier themes, while Italian films feature more domestic comedy and worker-related content. Notably, DMR reveals a "Factories, bosses & workers" topic (top words: factory, boss, engineer, priest, worker) that is 16 percentage points more prevalent in Italian films – a theme largely absent from American cinema that maps directly onto left-wing political discourse about labor and class. Table A3 presents the full DMR topic results.

Table A3: Full Topic List from DMR Model ($K = 10$)

#	Label	Top Words	IT	US	IT–US
0	Naval/sea combat	ship, boat, captain, aye, island, sea, miles, plane	0.019	0.048	−0.03
1	Royal & biblical epics	king, thee, slave, master, thy, fear, honor, power	0.094	0.075	+0.02
2	Wilderness/rural	dog, church, tree, dying, teach, smell, village	0.136	0.141	−0.00
3	Westerns: guns & gold	ain't, gold, dollars, horses, guns, sheriff, train	0.039	0.124	−0.08
4	Trials & justice	court, murder, guilty, state, witness, crime, trial	0.143	0.104	+0.04
5	Romance & showbiz	music, marriage, bye, sing, hotel, wedding, song	0.176	0.117	+0.06
6	Urban crime & hustlers	honey, fix, relax, kidding, million, apartment	0.078	0.159	−0.08
7	British manners	book, tea, pardon, fellow, absolutely, perfectly	0.080	0.160	−0.08
8	Army & war	army, general, soldier, officer, colonel, sergeant	0.047	0.054	−0.01
9	Factories & workers	million, boss, engineer, priest, factory, workers, professor	0.187	0.019	+0.17

Notes: Topics from Dirichlet Multinomial Regression with $K = 10$ on masked subtitle text, with country of origin and year as covariates. "IT" and "US" columns show mean topic share for Italian-origin and U.S.-origin films. "IT–US" is the difference in prevalence. The "Factories & workers" topic (row 9), with 16 percentage points higher prevalence in Italian films, includes themes of labor and class that map onto left-wing discourse.

C Persuasion Rate of U.S. Soft Power

To benchmark the magnitude of our electoral findings against the broader media economics literature, we adapt the persuasion rate framework formalized by DellaVigna and Gentzkow (2010). The standard persuasion rate, f, captures the fraction of an audience that alters their behavior upon exposure to a message, conditional on not already being predisposed to that behavior.

In a setting with a binary treatment group (T) and control group (C), the formula is typically expressed as:

$$f = \frac{y_T - y_C}{e_T - e_C} \times \frac{1}{1 - y_0} = \frac{\Delta y}{\Delta e} \times \frac{1}{1 - y_0} \tag{4}$$

where Δy is the treatment effect on the outcome (the intention-to-treat effect), Δe is the change in the share of the population actually exposed to the message, and y_0 is the baseline share of the population already engaging in the behavior. In canonical studies of media persuasion, such as the rollout of Fox News across U.S. towns, the treatment is a coarse geographic binary (e.g. whether the channel is available in a given location). Consequently, Δe is required to scale the geographic intention-to-treat effect down to the individual treatment-on-the-treated effect, operating largely on the extensive margin of new viewership.

Our historical setting and empirical strategy depart from this standard framework in two key ways. First, exposure to Hollywood cinema in post-war Italy was ubiquitous; the average Italian went to the movies more than 1.25 times a month (see Section 1). Thus, our treatment operates dynamically on the intensive margin (a dose-response) rather than solely on the extensive margin. Second, our exposure variable – U.S. movie tickets sold per capita, proxied in the main specification by the interaction of U.S. movie market share and local movie theater capacity or box office – is a continuous measure that inherently captures the density of exposure within a geographic unit. Because our measure directly quantifies treatment intensity, we do not require a separate Δe denominator to scale down a coarse geographic instrument.

To map our continuous treatment into a probability of persuasion, we calculate a *per-ticket* persuasion rate. We define the treatment dose as a single U.S. movie viewing. Therefore, a 1-unit increase in our raw exposure measure corresponds to exactly one additional U.S. movie ticket per capita ($\Delta e = 1$).

We recover the raw treatment effect (Δy) by dividing the standardized OLS coefficient from our tickets specification ($\beta_{std} = 0.00820$, Table 4, column (7)) by the standard deviation of the raw tickets exposure measure, i.e., the unstandardized interaction between the U.S. revenue share and tickets sold per capita ($SD_{raw} = 6.5511$). This yields a raw coefficient of $\beta_{raw} \approx 0.00125$. The baseline vote share for the DC party is $y_0 = 0.377$ (Table 1), meaning the persuadable share of the electorate is $(1-0.377) = 0.623$.

Applying the formula:

$$f_{per\_ticket} = \frac{0.00125}{1} \times \frac{1}{0.623} \approx 0.0020 \tag{5}$$

We estimate a baseline OLS per-ticket persuasion rate of **0.20%**. Applying the same calculation to the tickets-based IV estimates of Table 4 yields per-ticket rates of **0.18%** with the France instrument ($\beta_{std} = 0.00725$, column (9)) and **0.32%** with the Netherlands instrument ($\beta_{std} = 0.0132$, column (11)). Note that the per-ticket framework requires the tickets-based exposure measure, so these rates are not computed from our main availability-based specifications (Tables 2 and 3). To contextualize these increments, the average Italian in our sample purchased approximately 15.47 total movie tickets annually, with U.S. movies accounting for 39.65% of revenues (Table 1). This implies a baseline exposure of roughly 6.13 U.S. tickets per capita. An additional ticket therefore represents a 16.3% increase over baseline exposure.

Table A4 reproduces the summary of media persuasion rates from DellaVigna and Gentzkow (2010). Direct political persuasion strategies, such as the introduction of Fox News or anti-Putin television, yield persuasion rates ranging from 7.7% to 11.6%. Comparing a single movie viewing to the persistent availability of a partisan news network, however, requires accounting for treatment dosage. If we scale our per-ticket rates to represent an average voter's annual diet of American cinema (6.13 U.S. tickets), the annualized persuasion rate is approximately 1.2% for the OLS baseline and ranges from 1.1% (France instrument) to 2.0% (Netherlands instrument) across the IV estimates.

While more comparable, this annualized rate remains an order of magnitude smaller than direct persuasion strategies. This discrepancy perfectly highlights the theoretical distinction between overt propaganda and "soft power." The political messaging embedded in Hollywood cinema is subtle, indirect, and heavily diluted by entertainment value. Yet, what soft power lacks in individual persuasion, it makes up for in population reach. For example, while the rollout of Fox News yielded a high persuasion rate of 11.6%, its newly exposed audience was relatively small, ultimately shifting the aggregate town-level Republican vote share by roughly 0.15 percentage points (DellaVigna and Kaplan, 2007). In contrast, because Cold War Italians consumed a massive, nearly universal volume of U.S. movies, our small per-viewing persuasion rate can compound into substantial aggregate electoral shifts. A one-standard-deviation increase in our exposure index shifted the aggregate DC vote share by 0.73 to 1.32 percentage points across the tickets-based specifications of Table 4 (columns (7), (9), and (11)), demonstrating the quiet efficacy of cultural exports in shaping geopolitical alignment.

Table A4: Persuasion Rates: Summary of Studies (DellaVigna and Gentzkow, 2010)

Paper	**Treatment**	**Context**	**Variable y**	**Year**	**Time horizon**	**Persuasion f**
Persuading voters						
Gosnell 1926 (FE)	Registration card	Pres. election	Registration	1924	Few days	13.4%
Gerber & Green 2000 (FE)	Door-to-door GOTV	Cong. election	Turnout	1998	Few days	15.6%
Gerber & Green 2000 (FE)	1-3 mailings of GOTV cards	Cong. election	Turnout	1998	Few days	1.0%
Green et al. 2008 (FE)	Door-to-door canvassing	Local election	Turnout	2001	Few days	11.5%
Green & Gerber 2001 (FE)	Phone calls by 18-30-year-olds	Gen. election	Turnout	2000	Few days	20.4%
Green & Gerber 2001 (FE)	Phone calls	Gen. election	Turnout	2000	Few days	4.5%
DellaVigna & Kaplan 2007 (NE)	Fox news via cable	Pres. election	Rep. vote share	1996-00	0-4 years	11.6%
Enikolopov et al. 2010 (NE)	Anti-Putin TV (NTV)	Gen. election	Anti-Putin vote	1999	3 months	7.7%
Chiang & Knight 2009 (NE)	Unsurprising Dem endor. (NYT)	Gen. election	Support for Gore	2000	Few weeks	2.0%
Chiang & Knight 2009 (NE)	Surprising Dem endor. (Denver)	Gen. election	Support for Gore	2000	Few weeks	6.5%
Gerber et al. 2009 (FE)	Free WP subscription	Gov. election	Dem. vote share	2005	2 months	19.5%
Gentzkow 2006 (NE)	Exposure to TV	Cong. election	Turnout	1940-50	10 years	4.4%
Gentzkow & Shapiro 2009 (NE)	Read local newspaper	Pres. election	Turnout	1869-1928	0-4 years	12.9%

D Data Appendix

We construct several variables from the ITANES survey data to measure individuals' sentiments towards the U.S., Russia, the DC, and the PCI:

- One question asks respondents to evaluate the importance of the *"independence of Italy from the United States"* on a scale from 1 to 4 (with 1 indicating very important and 4 not at all important). We construct the dummy variable *Independence* which is equal to one whenever the individual answers that it is either of little importance or not at all important;

- Another question asks whether they agree or disagree with the statement that *"the foreign policy of Italy ought to be completely independent of the United States"*. We construct the dummy variable *Foreign Policy* whenever individuals disagree with the statement.

- Another question states: "Now we would like to know what you think of certain groups or organizations that have an influence on political life. Please give a score to them from 0 to 100 depending on how you feel toward them." It then indicates 100 as *Very Favorable* and 0 as *Very Unfavorable*. Among the categories of groups included there are "Americans", "Russians", the "DC", and the "PCI". From this we construct a variable - *US-USSR* - to measure whether the individual likes Americans or Russians more, we subtract the score for Russians from the score assigned for Americans and we then standardize the variable. We code the variable *DC-PCI* analogously using the scores from the DC and the PCI.

E Additional Tables and Figures

Table A5: National Elections: Results

Election	DC	PCI	3rd Party	3rd Party Name
1953	40.1	22.6	12.7[a]	Italian Socialist Party (PSI)
1958	42.4	22.7	14.2[a]	Italian Socialist Party (PSI)
1963	38.3	25.3	13.8[a]	Italian Socialist Party (PSI)
1968	39.1	26.9	14.5[b]	Unified Socialist Party (PSU)
1972	38.7	27.1	9.6[a]	Italian Socialist Party (PSI)

Notes: Vote shares of top three Italian parties from 1953, 1958, 1963, 1968, and 1972 national elections.

Table A6: Summary Statistics by Province Capital and Rest of the Province

	Province Capital			Rest of the Province		
	(1) Mean	(2) Std. Dev	(3) N	(4) Mean	(5) Std. Dev	(6) N
Cinema variables						
Movie theaters per 10,000 inhabitants	1.77	0.68	1869	2.30	0.90	1869
Number of movie theaters	26.32	40.29	1869	85.71	61.55	1869
Number of movie theaters in 1948	18.99	30.11	89	64.19	46.20	89
Tickets sold per capita (yearly)	20.67	7.23	1869	10.27	4.68	1869
Electoral variables						
DC Share	34.89	7.88	445	40.54	9.78	445
PCI Share	24.28	10.07	445	26.34	11.08	445
Turnout	95.25	3.30	445	92.29	4.90	445
Census variables						
Total Population	167823	302819	1869	379828	230333	1869
% employed in the agricultural sector	12.84	11.30	267	40.81	18.26	267
% employed in the industrial sector	36.75	8.34	267	36.08	13.61	267
% employed in the commercial sector	16.38	4.24	267	9.58	3.41	267
% employed in the services sector	34.03	8.59	267	13.54	5.08	267
Employment rate	43.85	4.79	267	48.44	5.71	267
Labor force participation	46.43	4.56	267	50.32	5.67	267
% with a college degree	10.35	3.16	267	3.78	1.62	267

Notes: Electoral variables for national elections of 1953, 1958, 1963, 1968, and 1972. Census variables are obtained from Italian Decennial Censuses (1951, 1961, and 1971). Variables are averaged over the period 1950-1973. Columns 1-3 report values for the sample of Province Capitals, while columns 4-6 report values for the Rest of the Provinces.

Table A7: Robustness to accounting for spatial correlation

	DC Share				PCI Share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% DC	% DC	% DC	% DC	% PCI	% PCI	% PCI	% PCI
Exposure	0.00751***	0.00751***	0.00751**	0.00751**	-0.00447**	-0.00447**	-0.00447**	-0.00447**
	(0.00287)	(0.00291)	(0.00296)	(0.00303)	(0.00216)	(0.00213)	(0.00213)	(0.00206)
Distance	10	20	30	50	10	20	30	50
N	890	890	890	890	890	890	890	890

Notes: Estimation of Model 1 using Conley standard errors with different bandwidths. All specifications include election year and location fixed effects. The dependent variable is the vote share of various parties as indicated in the heading of the column. The row Distance indicates the different bandwidths in kilometers used. All regressions include controls for sectoral employment, labor participation, population, and education. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A8: Robustness to controlling for location-specific time trends

	DC Share				PCI Share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% DC	% DC	% DC	% DC	% PCI	% PCI	% PCI	% PCI
Exposure	0.0122***	0.0130***	0.0130***	0.0134***	0.00326	0.00326	0.00312	0.00349
	(0.00424)	(0.00406)	(0.00407)	(0.00410)	(0.00243)	(0.00243)	(0.00242)	(0.00237)
Location-spec. Time Trend	✓	✓	✓	✓	✓	✓	✓	✓
Sectorial employment		✓	✓	✓		✓	✓	✓
Labor participation		✓	✓	✓		✓	✓	✓
Population			✓	✓			✓	✓
Education				✓				✓
Mean Dep. Variable	0.377				0.253			
SD Dep. Variable	0.0932				0.106			
Within R-sq	0.0264	0.0630	0.0630	0.0718	0.00423	0.0230	0.0254	0.0375
Locations	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890

Notes: OLS estimates from equation 1 in all columns. All specifications include election year, location fixed effects, and location-specific time trends (i.e., we estimate a linear trend in time for each geographical unit). The dependent variable is the vote share for the DC in columns (1)-(4) and for the PCI in columns (5)-(8). The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. Sectoral employment controls include controls for the share of people employed in the agricultural, industrial, commercial, and service sectors; labor participation controls include employment rate and labor force participation; population is the population logarithm; education indicates the share of people with a college degree. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A9: Robustness to including exposure measured in other years

	DC Share					PCI Share				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	% DC	% DC	% DC	% DC	% DC	% PCI	% PCI	% PCI	% PCI	% PCI
Exposure	0.00824***		0.00921**		0.00959**	-0.00495**		-0.00450		-0.00476
	(0.00296)		(0.00409)		(0.00413)	(0.00230)		(0.00367)		(0.00364)
Exposure (2 years)	-0.00150	-0.000412			-0.00142	0.000983	0.000333			0.000993
	(0.00169)	(0.00168)			(0.00171)	(0.00114)	(0.00116)			(0.00117)
Exposure (Election year)			-0.00232	0.00450	-0.00189			0.0000434	-0.00329	-0.000256
			(0.00426)	(0.00308)	(0.00433)			(0.00396)	(0.00254)	(0.00406)
Within R-sq	0.123	0.112	0.122	0.115	0.124	0.135	0.128	0.134	0.131	0.135
Locations	178	178	178	178	178	178	178	178	178	178
N	890	890	890	890	890	890	890	890	890	890

Notes: The main exposure measure is measured in the year prior to the election and the related coefficient is reported in the top row (see discussion in Section 4). We run a "horse race" regression with exposure measured two years prior in columns (1) and (6). We do the same using exposure measured in the election year in columns (3) and (8). Columns (5) and (10) display the results from a regression including exposure in the same year of the election, in the year prior (our preferred exposure measure, see discussion in Section 4), and two years prior. All specifications include election year and location fixed effects. All regressions include controls for sectoral employment, labor participation, population, and education. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A10: Results on PCI membership

	PCI Branches per 1,000 inhabitants				PCI Members per 1,000 inhabitants			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Exposure	-0.00116	-0.000479	-0.000428	-0.000384	0.211	0.334	0.366	0.364
	(0.00264)	(0.00254)	(0.00256)	(0.00251)	(0.741)	(0.694)	(0.689)	(0.688)
Sectorial employment		✓	✓	✓		✓	✓	✓
Labor participation		✓	✓	✓		✓	✓	✓
Population			✓	✓			✓	✓
Education				✓				✓
Mean Dep. Variable	0.224				39.97			
SD Dep. Variable	0.128				35.71			
Locations	89	89	89	89	89	89	89	89
N	1596	1596	1596	1596	1593	1593	1593	1593

Notes: The table reports results on the number of PCI branches per 1,000 inhabitants in columns (1)-(4) and for the number of PCI members per 1,000 inhabitants in columns (5)-(8). The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. Sectoral employment controls include controls for the share of people employed in the agricultural, industrial, commercial, and service sectors; labor participation controls include employment rate and labor force participation; population indicates the logarithm of the resident population; and education indicates the share of people with a college degree. All specifications include election year and location fixed effects. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A11: Turnout: incidence and electoral accounting

Panel A: Incidence of the turnout effect (dep. var.: turnout)				
	All areas	PCI-leaning areas	DC-leaning areas	Interaction model
Exposure	-0.00426**	-0.00376**	-0.00567	-0.00507**
	(0.00189)	(0.00173)	(0.00375)	(0.00197)
Exposure $\times$ DC-leaning				0.00164
				(0.00153)
Mean Dep. Variable	0.938	0.949	0.927	0.938
Locations	178	89	89	178
N	890	445	445	890
Panel B: Electoral accounting (dep. var.: share of eligible electorate)				
	DC	PCI	Other	Abstention
Exposure	0.00545*	-0.00604***	-0.00368	0.00426**
	(0.00282)	(0.00231)	(0.00331)	(0.00189)
Mean Dep. Variable	0.352	0.238	0.347	0.0623
Locations	178	178	178	178
N	890	890	890	890

Notes: OLS estimates. All specifications include election year and location fixed effects and the full set of controls (sectoral employment, labor participation, population, and education). The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. Panel A reports the effect on turnout for all areas, for PCI-leaning and DC-leaning areas separately (split at the 1948 DC vote share, as in Table A12), and an interaction model in which Exposure is interacted with an indicator for DC-leaning areas. Panel B decomposes the eligible electorate into the vote shares of the DC, the PCI, all other outcomes (other parties, blank, and invalid ballots), and abstention, each expressed as a share of the eligible electorate; the four columns partition the electorate, so the coefficients sum to zero. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A12: Heterogeneity: DC vs PCI Areas

Panel a: DC Areas

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00492	0.0107	-0.00606**	-0.0208*
	(0.00492)	(0.0121)	(0.00289)	(0.0121)
Method	OLS	IV	OLS	IV
Locations	89	89	89	89
N	445	445	445	445

Panel b: PCI Areas

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00954***	0.0342***	-0.00379	-0.0313**
	(0.00344)	(0.0109)	(0.00347)	(0.0125)
Method	OLS	IV	OLS	IV
Locations	89	89	89	89
N	445	445	445	445

Notes: OLS estimates from equation 1 in all columns. All specifications include election year and location fixed effects. The top panel reports results for DC areas and the bottom panel reports results for PCI areas. The dependent variable is the vote share of the party as indicated in the heading of the column. The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. All regressions include controls for sectoral employment, labor participation, population, and education. The row Method indicates whether we are estimating equation with OLS or IV. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A13: Heterogeneity: Province Capitals vs Rest of the Province

Panel a: Province Capitals

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00901**	0.0232*	-0.00235	-0.00787
	(0.00355)	(0.0128)	(0.00288)	(0.0153)
Method	OLS	IV	OLS	IV
Locations	89	89	89	89
N	445	445	445	445

Panel b: Rest of the Province

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00294	0.0253**	-0.0108***	-0.0439***
	(0.00423)	(0.0121)	(0.00353)	(0.0117)
Method	OLS	IV	OLS	IV
Locations	89	89	89	89
N	445	445	445	445

Notes: OLS estimates from equation 1 in all columns. All specifications include election year and location fixed effects. The top panel reports results for Province Capitals and the bottom panel reports results for the Rest of the Province. The dependent variable is the vote share of the party as indicated in the heading of the column. The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. All regressions include controls for sectoral employment, labor participation, population, and education. The Method row indicates whether we are estimating equation with OLS or IV. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Table A14: Heterogeneity by geographical area

Panel a: North

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	-0.00232	-0.0195	-0.00510	-0.00938
	(0.00334)	(0.0124)	(0.00309)	(0.0158)
Method	OLS	IV	OLS	IV
Locations	74	74	74	74
N	370	370	370	370

Panel b: Center

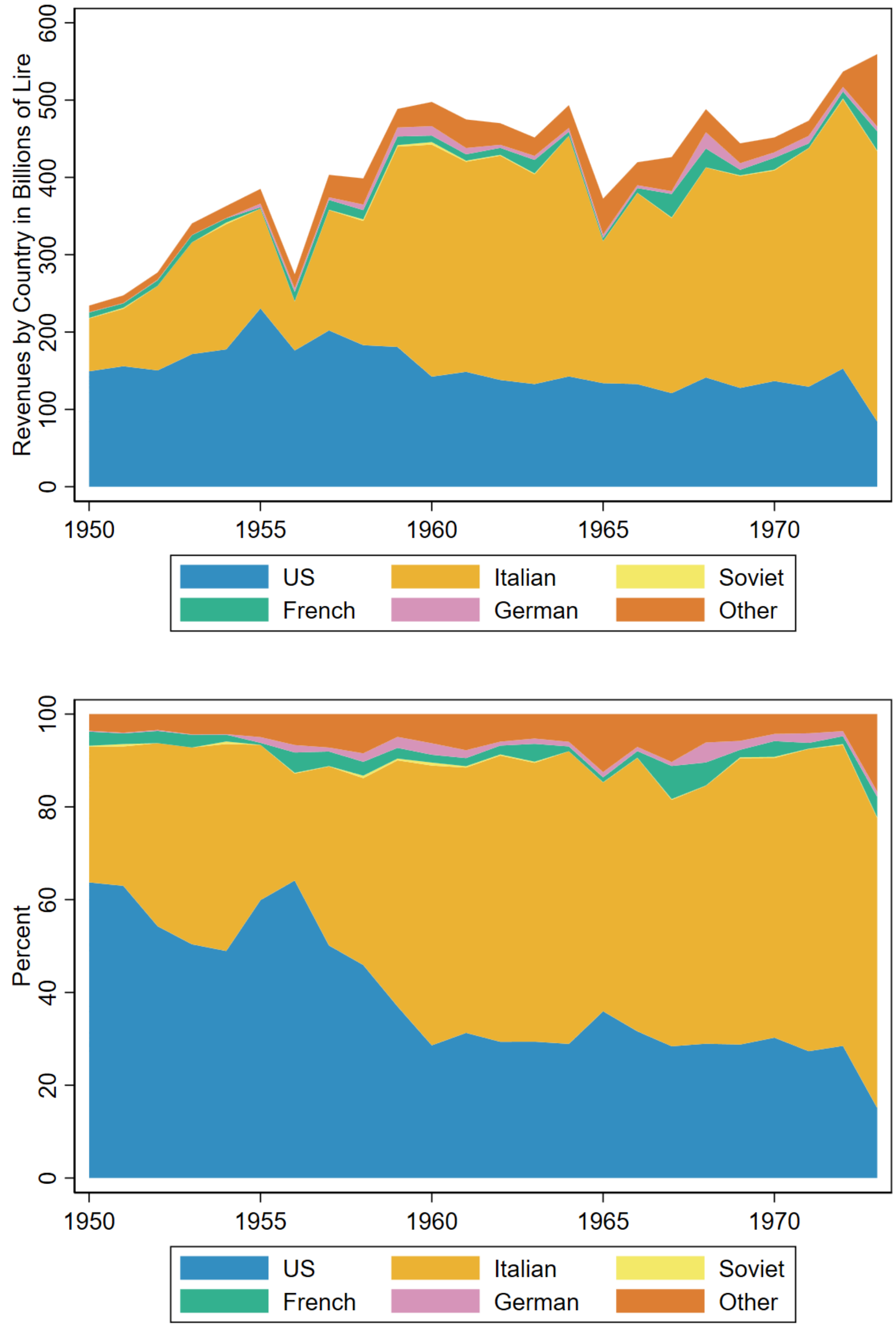
	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00534	0.00648	0.00547	0.0102
	(0.00364)	(0.00786)	(0.00391)	(0.0122)
Method	OLS	IV	OLS	IV
Locations	40	40	40	40
N	200	200	200	200

Panel c: South and Islands

	DC Share		PCI Share	
	(1)	(2)	(3)	(4)
Exposure	0.00297	0.000928	-0.00356	-0.0280***
	(0.00326)	(0.0130)	(0.00344)	(0.0105)
Method	OLS	IV	OLS	IV
Locations	64	64	64	64
N	320	320	320	320

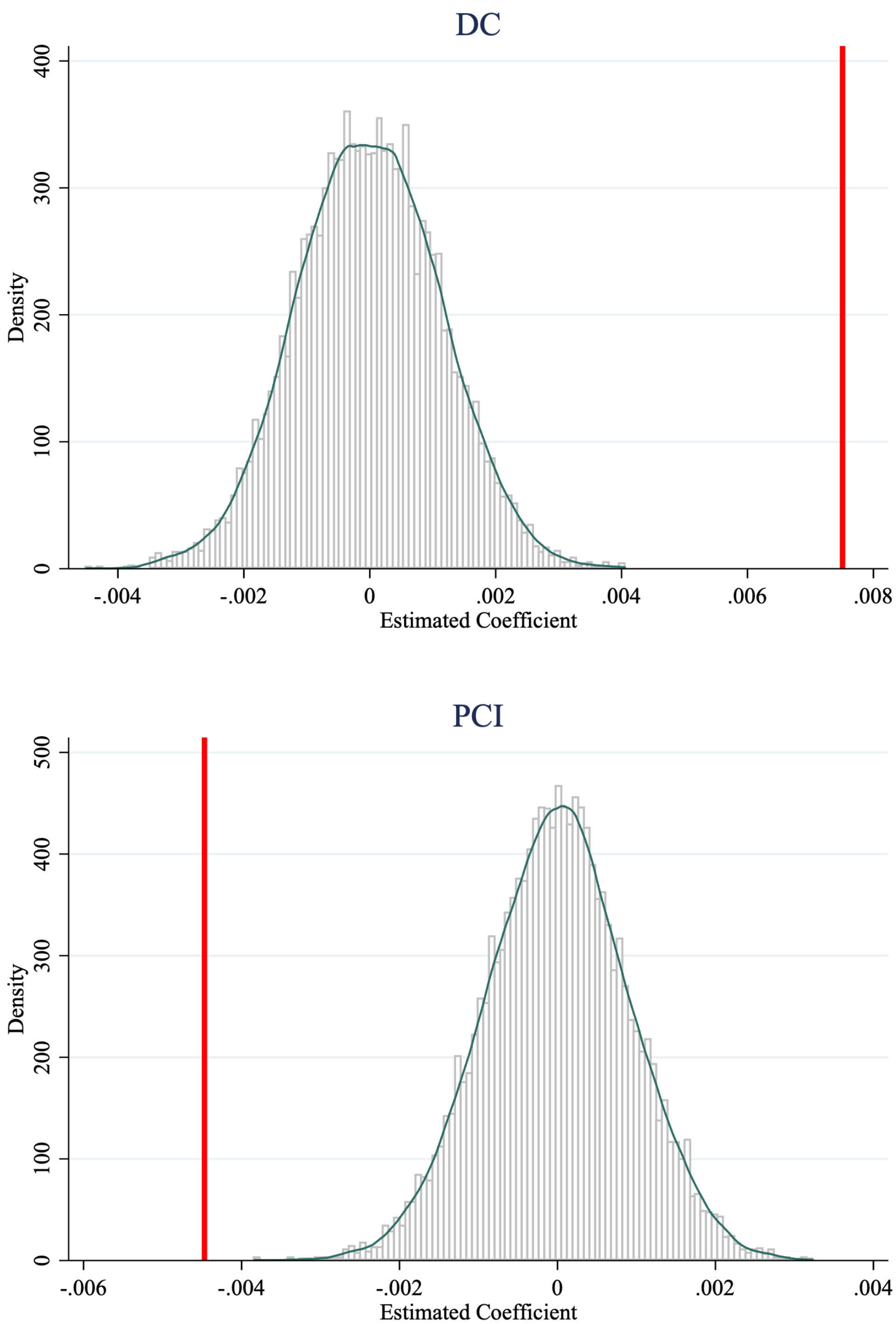
Notes: OLS estimates from equation 1 in all columns. All specifications include election year and location fixed effects. Panel A reports results for Northern Italy, Panel B reports results from Central Italy, and Panel C reports results from Southern Italy. The dependent variable is the vote share of the party as indicated in the heading of the column. The independent variable, Exposure, is the interaction term between local exposure to cinema and the success of U.S. movies as explained in Section 4, and it is standardized. All regressions include controls for sectoral employment, labor participation, population, and education. The row Method indicates whether we are estimating equation with OLS or IV. Standard Errors are clustered at the location level. * denotes significance at 10%, ** significance at 5% and *** significance at 1%.

Figure A1: Composition of Movie Revenues over Time



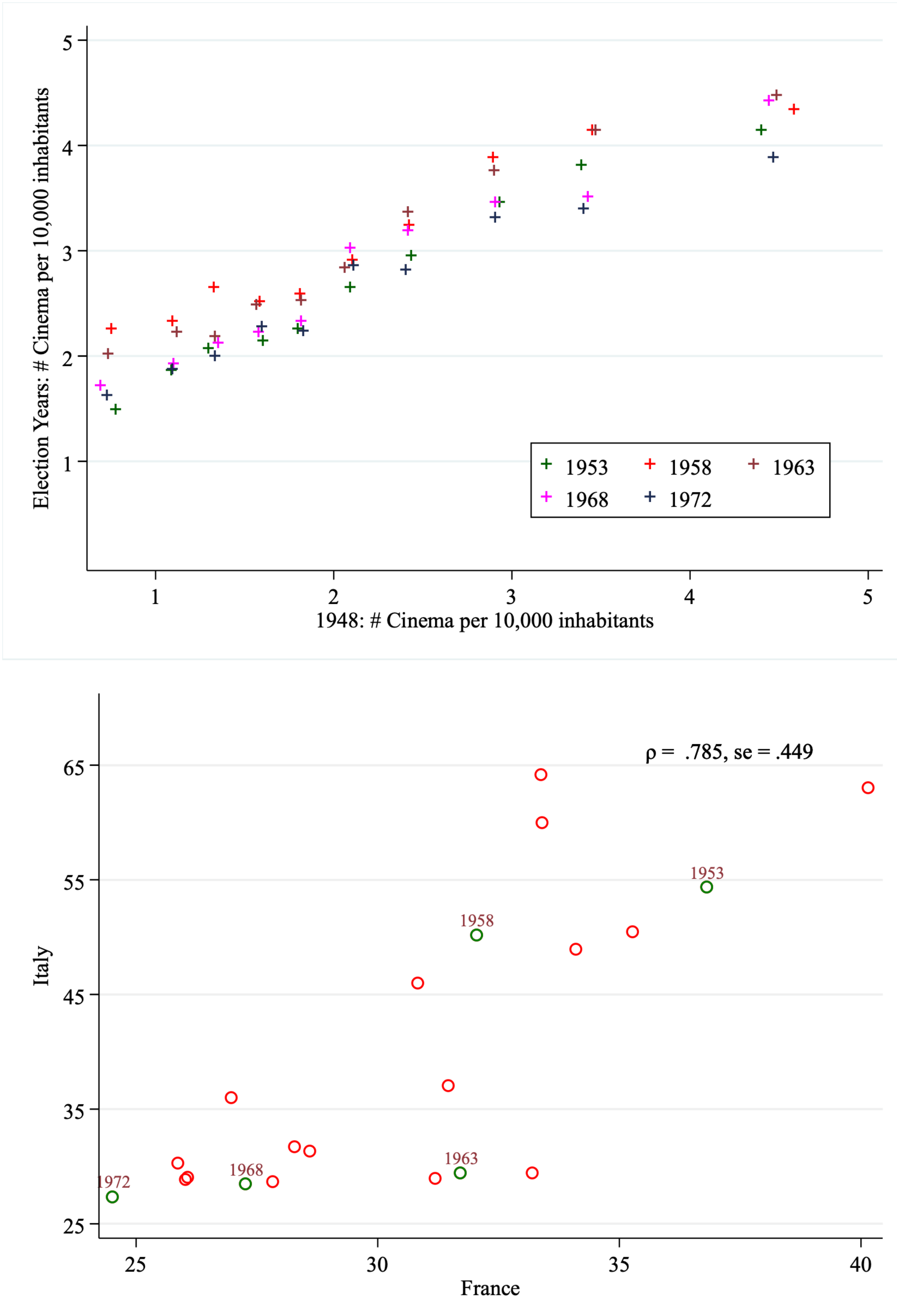
Notes: The upper panel displays total movie revenues by country of origin in billions of Lire; the lower panel displays the same data as a share of total revenues. Data on movie revenues by year and country of origin from SIAE's yearly reports, "Annuari dello Spettacolo." Italian and American movies represent over three quarters of total revenues in every year of the relevant period, while the revenue share of Soviet movies is always below 1%.

Figure A2: Falsification Test



Notes: Both Figures show a histogram of estimated coefficients from equation (1), based on 10,000 randomly shuffled years. The vertical line indicates the point estimate from columns 4 (top panel) and column 8 (bottom panel) from Table 2 .

Figure A3: Partial First Stage



Notes: the Figure at the top is a binscatter displaying the relationship between the number of movie theaters per 10,000 inhabitants in 1948 and subsequent national election years. The Figure at the bottom plots the relationship between the share of total movie revenues generated by U.S. movies in France and in Italy.